

ARVIND LIMITED

Quality Inspection Tracker

Architecture and complete source listing for the Full Stack Developer hiring assignment.

Assignment	Full Stack Developer
Frontend	React 19 · Vite · TypeScript · Tailwind CSS
Backend	Node 24 · Express 5 · Prisma 7 · PostgreSQL 16
Deployment	Docker · docker compose · nginx
Generated	21 August 2026

73

FILES

3,999

LINES

9

ENDPOINTS

3

CONTAINERS

Contents

1 • Architecture

overview, data model, API

- | | |
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| 1.1 System overview | 1.5 Backend structure |
| 1.2 Request lifecycle | 1.6 Frontend structure |
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2 • Project root

5 files

README.md	.gitignore
docker-compose.yml	LICENSE
.env.example	

3 • Backend — apps/api

31 files · paths relative to apps/api/

.env.example	src/app.ts
.gitignore	src/env.ts
package.json	src/prisma.ts
tsconfig.json	src/seed.ts
tsconfig.build.json	src/lib/asyncHandler.ts
eslint.config.js	src/lib/dates.ts
.prettierrc.json	src/lib/errors.ts
.prettierignore	src/middleware/auth.ts
.dockerignore	src/middleware/errorHandler.ts
Dockerfile	src/modules/auth/auth.routes.ts
docker-entrypoint.sh	src/modules/inspections/inspection.mapper.ts
prisma.config.ts	src/modules/inspections/inspection.routes.ts
prisma/schema.prisma	src/modules/inspections/inspection.schemas.ts
prisma/migrations/20260820122703_init/migration.sql	src/modules/inspections/inspection.service.ts
prisma/migrations/migration_lock.toml	src/modules/sap/sap.routes.ts
src/server.ts	

4 • Frontend — apps/web

37 files · paths relative to apps/web/

.env.example	src/components/AppLayout.tsx
.gitignore	src/components/FilterSheet.tsx
package.json	src/components/InspectionCard.tsx
tsconfig.json	src/context/AuthContextValue.ts
tsconfig.app.json	src/context/AuthProvider.tsx
tsconfig.node.json	src/hooks/useAuth.ts
eslint.config.js	src/hooks/useInspections.ts
.prettierrc.json	src/lib/api.ts
.prettierignore	src/lib/cn.ts
.dockerignore	src/lib/format.ts
Dockerfile	src/lib/labels.ts
nginx.conf	src/lib/types.ts
vite.config.ts	src/pages/InspectionDetailPage.tsx
index.html	src/pages/ListPage.tsx
src/main.tsx	src/pages/LoginPage.tsx
src/App.tsx	src/pages/NewInspectionPage.tsx
src/index.css	src/pages/SummaryPage.tsx
public/favicon.svg	src/components/ui/index.tsx
src/vite-env.d.ts	

Generated lockfiles (`package-lock.json` , 8,128 lines) are omitted. Every other tracked file is reproduced in full.

1 · Architecture

1.1 System overview

Three containers behind one compose file. The browser only ever talks to nginx, which serves the built single-page app and forwards `/api` to the Express service. Because both come from the same origin there is no CORS configuration and no API host baked into the JavaScript bundle.



1.2 Request lifecycle

Every write follows the same path, and each stage has one job:

STAGE	RESPONSIBILITY
nginx	Serves the SPA; proxies <code>/api</code> to the API container.
Middleware	helmet, CORS, JSON body limit (256 kb), request logging.
requireAuth	Verifies the JWT. Rejects with 401 before any handler runs.
Route	Parses input with Zod, picks a status code. No business logic.
Service	Normalises input, applies the rules, talks to Prisma.
Mapper	Converts the row to a DTO — dates become strings here.
Error handler	Turns every throw into the one error envelope.

1.3 Data model

Two tables. `inspections` holds the domain; `users` exists only for the auth bonus. Defect type, severity, status and source are PostgreSQL enums rather than lookup tables, so the database rejects an invalid value even if it somehow bypasses validation.

COLUMN	TYPE	NOTE
id	uuid	Primary key
inspection_date	DATE	Calendar day, pinned to UTC — no timezone drift
machine_id	varchar(64)	Free text, trimmed and uppercased on write
defect_type	enum	WEAVE_DEFECT · SHADE_VARIATION · HOLE_TEAR · COUNT_DEVIATION · OTHER
severity	enum	CRITICAL · MAJOR · MINOR — declared in urgency order
remarks	varchar(2000)	Optional
status	enum	OPEN · RESOLVED

resolution_note	varchar(2000)	Required to move to RESOLVED
resolved_at	timestamp	Set when resolved
source	enum	MANUAL · SAP_WEBHOOK
client_ref	varchar(64) unique	Idempotency key for SAP replays
sap_payload	jsonb	Raw webhook envelope, kept for traceability
created_at / updated_at	timestamp	Audit fields

Indexes: `(status, severity)` for the filtered list, `inspection_date DESC` and `created_at DESC` for the default sorts, plus the unique index on `client_ref`.

1.4 API surface

METHOD	PATH	AUTH	PURPOSE
GET	/api/health	—	Liveness, used by the compose healthcheck
POST	/api/auth/login	—	Returns a JWT
GET	/api/auth/me	JWT	Validates a stored token on boot
GET	/api/inspections	JWT	Filter, sort, paginate
POST	/api/inspections	JWT	Create
GET	/api/inspections/:id	JWT	Single record
PATCH	/api/inspections/:id/resolve	JWT	Resolve with a mandatory note
GET	/api/inspections/summary	JWT	Counts by severity and status
POST	/api/sap-webhook	x-sap-signature	Creates an inspection from an SAP payload

1.5 Backend structure

Feature modules, each with the same four parts, so a reader can predict where anything lives. Route handlers never touch Prisma.

FILE	HOLDS
*.schemas.ts	Zod schemas — the only place input shape is defined
*.service.ts	Business rules and all Prisma calls
*.routes.ts	Thin HTTP layer: parse, delegate, choose a status code
*.mapper.ts	Database row → API DTO

1.6 Frontend structure

Pages compose components; components stay presentational; every server call goes through one typed API client. TanStack Query owns caching and refetching, so no page holds server data in local state.

FOLDER	HOLDS
pages/	One file per screen: list, new, detail, summary, login
components/	Layout, inspection card, filter sheet, and the UI primitives
hooks/	Query and mutation hooks, plus useAuth
lib/	API client, formatting, labels, shared types
context/	Auth provider and its context object

1.7 Deployment

The API image is multi-stage: dependencies and the Prisma client are built in a Node stage, then only `dist`, `prisma` and production dependencies are copied into a slim runtime that runs as a non-root user. On boot the entrypoint verifies the generated client is present, applies migrations, seeds demo data on an empty database, and then starts the server — so `docker compose up --build` is the only command a reviewer needs.

2 · Project root

Compose file, environment template and repository config.

README.md

MD · 143 LINES

```
1 # Quality Inspection Tracker
2
3 A mobile-first web app for Arvind shop-floor supervisors to log, track and resolve fabric quality defects –
  replacing the paper register and the spreadsheet re-entry that follows it.
4
5 **Stack:** React 19 · Vite · TypeScript · Tailwind CSS · Node · Express 5 · Prisma 7 · PostgreSQL 16 ·
  Docker
6
7 ---
8
9 ## Setup
10
11 ### With Docker
12
13 ```bash
14 git clone https://github.com/HARDIK-31/quality-inspection-tracker.git
15 cd quality-inspection-tracker
16 docker compose up --build
17 ```
18
19 | | |
20 |---|---|
21 | App | <http://localhost:8080> |
22 | API | <http://localhost:4000/api/health> |
23 | Login | `supervisor` / `arvind123` |
24
25 That is the only command. The API container applies migrations on start and seeds ~25 demo inspections
  across every severity, status and defect type, so the list, filters and summary all have data on first
  load.
26
27 ### Without Docker
28
29 Needs Node 20+ and a PostgreSQL 14+ instance.
30
31 ```bash
32 # API – terminal 1
33 cd apps/api
34 cp .env.example .env          # set DATABASE_URL
35 npm install
36 npx prisma generate           # client is generated, not committed
37 npx prisma migrate deploy
38 npm run db:seed
39 npm run dev                   # http://localhost:4000
40
41 # Web – terminal 2
42 cd apps/web
43 npm install
44 npm run dev                   # http://localhost:5173
45 ```
46
47 The Vite dev server proxies `/api` to port 4000, so there is no CORS setup and no API URL to configure.
  Every environment variable has a working default; see the `.env.example` files if you need to change one.
48
49 ---
50
51 ## What's implemented
52
53 **All four required features**
54
```

```

55 - **Log an inspection** - date (defaults to today, backdateable), machine/line ID, defect type, severity,
    optional remarks
56 - **List** - filter by severity, status, date range, defect type and machine ID; six sort orders; paginated
57 - **Resolve** - mandatory resolution note, enforced on both client and server
58 - **Summary** - open/resolved counts by severity, plus totals
59
60 **Two of the three bonuses**
61
62 - **Mock SAP integration** - `POST /api/sap-webhook`, secured with a shared-secret header ([details below]
    (#sap-webhook))
63 - **Authentication** - JWT with bcrypt-hashed passwords; login is required on every route except
    `/api/health` and the SAP webhook
64
65 **Not implemented**
66
67 - **Offline sync** - the app requires connectivity.
68
69 ---
70
71 ## Mobile-first
72
73 Designed and tested at **390px** in a real browser: no horizontal scrolling on any screen, every tap target
    at least 44px, and all text inputs at 16px so iOS Safari does not zoom on focus. Inspections render as
    cards with a coloured severity rail rather than table rows, which are unreadable at that width. Filters
    open in a bottom sheet so they cost no screen height while scanning the list.
74
75 ---
76
77 ## API
78
79 Base path `/api`. Errors share one shape:
80
81 | Method | Path | Notes |
82 |---|---|---|
83 | `GET` | `/health` | Liveness. No auth. |
84 | `POST` | `/auth/login` | `{ username, password }` → `{ token, user }` |
85 | `GET` | `/auth/me` | Validates a stored token |
86 | `GET` | `/inspections` | Filter, sort, paginate |
87 | `POST` | `/inspections` | `201` on success |
88 | `GET` | `/inspections/:id` | Single record |
89 | `PATCH` | `/inspections/:id/resolve` | `{ resolutionNote }` - `422` if blank, `409` if already resolved |
90 | `GET` | `/inspections/summary` | Counts by severity and status |
91 | `POST` | `/sap-webhook` | Requires `x-sap-signature`. No login. |
92
93 ### SAP webhook
94
95 SAP cannot complete an interactive login, so this endpoint sits outside the JWT wall and is authenticated
    with a **shared secret in the `x-sap-signature` header instead. A request with a missing or incorrect
    header is rejected with `401` before the payload is even parsed.
96
97 The secret comes from `SAP_WEBHOOK_SECRET` and defaults to `x-arvind-001-dhrq24sd3-sap`; override it in
    `.env` or the environment.
98
99 ```bash
100 curl -X POST http://localhost:4000/api/sap-webhook \
101   -H 'content-type: application/json' \
102   -H 'x-sap-signature: x-arvind-001-dhrq24sd3-sap' \
103   -d '{
104     "notification_no": "10000456",
105     "plant_code": "AHM-01",
106     "posting_date": "2026-08-19",
107     "work_center": "LOOM-14",
108     "defect_code": "WEAVE",
109     "priority": 1,
110     "long_text": "Auto-raised from SAP QM notification."
111   }'
112 ```
113 ---

```

```

114
115 ## Architecture decisions
116
117 **Two separate apps, one Docker setup.**
118 The API and the web app are independent npm packages, tied together only by `docker-compose.yml`. I skipped
monorepo tools because they add a layer that can break on someone else's machine, and these two never
import each other.
119
120 **Express, with the logic kept out of the routes.**
121 Nine plain endpoints, no realtime and no background jobs, so Express is enough. Routes only read the
request and pick a status code; the real work lives in separate service files.
122
123 **The same validation rules on both sides.**
124 Zod checks every request on the server, and the same rules run in the browser through react-hook-form. That
is duplicated on purpose: the browser copy gives instant feedback, the server copy is what actually
protects the database.
125
126 **Database enums, and a single table.**
127 The brief describes one thing – an inspection – so defect type, severity and status are Postgres enums
instead of extra lookup tables. Severity is declared Critical, Major, Minor, so sorting by that column is
already sorting by urgency.
128
129 **The inspection date is a plain date, not a timestamp.**
130 A supervisor records which day a defect was found, not the exact moment, and a timestamp would make an 11pm
entry in India show up as the previous day elsewhere. The created and resolved times are still full
timestamps, because there the moment does matter.
131
132 **nginx serves the app and forwards `/api` to the API.**
133 The browser uses one address for both, so there is no CORS setup and no API URL baked into the frontend
build. The dev server forwards the same way, so local development and Docker behave identically.
134
135 ---
136
137 ## What I'd do differently with more time
138
139 - **Automated tests.** The biggest gap – verified by hand instead. Vitest plus Supertest against a
throwaway Postgres would come first.
140 - **Offline support.** Needs an IndexedDB write queue *and* a service worker for the app shell; half of it
would silently lose entries.
141 - **Per-user attribution.** No `created_by` or `resolved_by`, so the audit trail is weaker than the paper
register it replaces.
142 - **Photo attachments.** The most useful missing field for a fabric defect; skipped because it pulls in
object storage.
143 - **Editing an inspection.** Only create and resolve exist, so a typo in a machine ID cannot be corrected.

```

docker-compose.yml

YAML · 68 LINES

```

1 # docker compose up --build
2 # app http://localhost:8080
3 # api http://localhost:4000/api/health
4
5 services:
6   postgres:
7     image: postgres:16-alpine
8     restart: unless-stopped
9
10    environment:
11      POSTGRES_USER: ${POSTGRES_USER:-qit}
12      POSTGRES_PASSWORD: ${POSTGRES_PASSWORD:-qit_ard}
13      POSTGRES_DB: ${POSTGRES_DB:-qit}
14    ports:
15      - '${POSTGRES_PORT:-5432}:5432'
16    volumes:
17      - postgres-data:/var/lib/postgresql/data
18    healthcheck:
19      test: ['CMD-SHELL', 'pg_isready -U ${POSTGRES_USER:-qit} -d ${POSTGRES_DB:-qit}']
20      interval: 5s
21      timeout: 5s

```



```

21     retries: 12
22     start_period: 10s
23
24     api:
25         build:
26             context: ./apps/api
27             dockerfile: Dockerfile
28             restart: unless-stopped
29             depends_on:
30                 postgres:
31                     condition: service_healthy
32             environment:
33                 NODE_ENV: production
34                 PORT: 4000
35                 DATABASE_URL: postgresql://${POSTGRES_USER:-qit}:${POSTGRES_PASSWORD:-
qit_ard}@postgres:5432/${POSTGRES_DB:-qit}?schema=public
36                 CORS_ORIGIN: ${CORS_ORIGIN:-*}
37                 JWT_SECRET: ${JWT_SECRET:-52139b6dc139f737a06bf90bb2ea63878f783cfdd038766d67bce5273c2748a}
38                 JWT_EXPIRES_IN: ${JWT_EXPIRES_IN:-12h}
39                 SAP_WEBHOOK_SECRET: ${SAP_WEBHOOK_SECRET:-x-arvind-001-dhrq24sd3-sap}
40                 # No-op once the database has rows.
41                 SEED_ON_START: ${SEED_ON_START:-true}
42             ports:
43                 - '${API_PORT:-4000}:4000'
44             healthcheck:
45                 test:
46                     [
47                         'CMD',
48                         'node',
49                         '-e',
50                         "fetch('http://127.0.0.1:4000/api/health').then(r=>process.exit(r.ok?
0:1)).catch(()=>process.exit(1))",
51                     ]
52                 interval: 10s
53                 timeout: 5s
54                 retries: 6
55                 start_period: 20s
56
57         web:
58             build:
59                 context: ./apps/web
60                 dockerfile: Dockerfile
61                 restart: unless-stopped
62                 depends_on:
63                     - api
64                 ports:
65                     - '${WEB_PORT:-8080}:80'
66
67     volumes:
68         postgres-data:

```

.env.example

ENV · 22 LINES

```

1  # --- Postgres -----
2  POSTGRES_USER=qit
3  POSTGRES_PASSWORD=qit_ard
4  POSTGRES_DB=qit
5  POSTGRES_PORT=5432
6
7  # --- Ports on the host -----
8  API_PORT=4000
9  WEB_PORT=8080
10
11 # --- API -----
12 CORS_ORIGIN=*
13
14 # Auth

```

```

15 JWT_SECRET=52139b6dc139f737a06bf90bb2ea63878f783cfcdd038766d67bce5273c2748a
16 JWT_EXPIRES_IN=12h
17
18 # Shared secret the SAP webhook expects in the x-sap-signature header.
19 SAP_WEBHOOK_SECRET=x-arvind-001-dhrq24sd3-sap
20
21 # Seeds demo inspections.
22 SEED_ON_START=true

```

.gitignore

IGNORE · 19 LINES

```

1 node_modules/
2 dist/
3 build/
4
5 # Prisma client is generated at build time.
6 apps/api/src/generated/
7
8 .env
9
10 .env.local
11
12 *.log
13 npm-debug.log*
14 coverage/
15
16 .DS_Store
17 .idea/
18 .vscode/
19 *.swp

```

LICENSE

TXT · 201 LINES

```

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201     limitations under the License.
```

3 · Backend — apps/api

Express 5 + Prisma 7 REST API, its Docker image and database schema.

apps/api/.env.example

ENV · 17 LINES

```
1 # Copy to `.env` for local (non-Docker) development.
2 # docker-compose supplies these itself.
3
4 NODE_ENV=development
5 PORT=4000
6
7 # Use `localhost` on the host, or `postgres` inside Docker.
8 DATABASE_URL="postgresql://qit:qit_ard@localhost:5432/qit?schema=public"
9
10 # Comma-separated allowlist, or * for any origin.
11 CORS_ORIGIN=*
12
13 JWT_SECRET=a9e3feaf15997f1c948d0f240a2382589a59aa98a0a6e262395feff3b11f8d36
14 JWT_EXPIRES_IN=12h
15
16 # Shared secret the SAP webhook expects in the x-sap-signature header.
17 SAP_WEBHOOK_SECRET=x-arvind-001-dhrq24sd3-sap
```

apps/api/.gitignore

IGNORE · 5 LINES

```
1 node_modules
2 # Keep environment variables out of version control
3 .env
4
5 /src/generated/prisma
```

apps/api/package.json

JSON · 52 LINES

```
1 {
2   "name": "qit-api",
3   "version": "1.0.0",
4   "private": true,
5   "type": "module",
6   "scripts": {
7     "dev": "tsx watch src/server.ts",
8     "build": "tsc -p tsconfig.build.json",
9     "start": "node dist/server.js",
10    "typecheck": "tsc --noEmit",
11    "lint": "eslint .",
12    "lint:fix": "eslint . --fix",
13    "format": "prettier --write \"src/**/*.ts\" \"prisma/**/*.ts\"",
14    "db:generate": "prisma generate",
15    "db:migrate": "prisma migrate dev",
16    "db:deploy": "prisma migrate deploy",
17    "db:seed": "tsx src/seed.ts",
18    "db:studio": "prisma studio",
19    "db:seed:prod": "node dist/seed.js",
20    "db:seed:force": "SEED_FORCE=true tsx src/seed.ts"
21  },
22  "dependencies": {
23    "@prisma/adapters-pg": "^7.9.1",
24    "@prisma/client": "^7.9.1",
25    "bcryptjs": "^3.0.3",
26    "cors": "^2.8.6",
27    "dotenv": "^17.4.2",
28    "express": "^5.2.1",
29    "helmet": "^8.3.0",
30    "jsonwebtoken": "^9.0.3",
```

```

31   "morgan": "^1.11.0",
32   "pg": "^8.23.0",
33   "prisma": "^7.9.1",
34   "zod": "^4.4.3"
35 },
36 "devDependencies": {
37   "@eslint/js": "^10.0.1",
38   "@types/bcryptjs": "^2.4.6",
39   "@types/cors": "^2.8.19",
40   "@types/express": "^5.0.6",
41   "@types/jsonwebtoken": "^9.0.10",
42   "@types/morgan": "^1.9.10",
43   "@types/node": "^26.2.0",
44   "@types/pg": "^8.23.1",
45   "eslint": "^10.8.1",
46   "eslint-config-prettier": "^10.1.8",
47   "prettier": "^3.9.6",
48   "tsx": "^4.23.12",
49   "typescript": "^6.0.3",
50   "typescript-eslint": "^8.67.0"
51 }
52 }

```

apps/api/tsconfig.json

JSON · 28 LINES

```

1 {
2   "compilerOptions": {
3     "target": "ES2023",
4     "lib": [
5       "ES2023"
6     ],
7     "module": "nodenext",
8     "moduleResolution": "nodenext",
9     "allowImportingTsExtensions": true,
10    "rewriteRelativeImportExtensions": true,
11    "verbatimModuleSyntax": true,
12    "strict": true,
13    "noUncheckedIndexedAccess": true,
14    "noImplicitOverride": true,
15    "noFallthroughCasesInSwitch": true,
16    "exactOptionalPropertyTypes": false,
17    "esModuleInterop": true,
18    "resolveJsonModule": true,
19    "skipLibCheck": true,
20    "forceConsistentCasingInFileNames": true,
21    "noEmit": true
22  },
23  "include": [
24    "src",
25    "prisma.config.ts",
26    "eslint.config.js"
27  ]
28 }

```

apps/api/tsconfig.build.json

JSON · 11 LINES

```

1 {
2   "extends": "../tsconfig.json",
3   "compilerOptions": {
4     "noEmit": false,
5     "rootDir": "src",
6     "outDir": "dist",
7     "sourceMap": true,
8     "declaration": true
9   },
10  "include": ["src"]

```

```
11 }
```

apps/api/eslint.config.js

TS · 24 LINES

```
1 import js from '@eslint/js';
2 import tseslint from 'typescript-eslint';
3 import prettier from 'eslint-config-prettier';
4
5 export default tseslint.config(
6   { ignores: ['dist', 'node_modules', 'src/generated'] },
7   js.configs.recommended,
8   ...tseslint.configs.recommendedTypeChecked,
9   {
10     languageOptions: {
11       parserOptions: { projectService: true, tsconfigRootDir: import.meta.dirname },
12     },
13     rules: {
14       '@typescript-eslint/no-unused-vars': ['error', { argsIgnorePattern: '^_' }],
15       '@typescript-eslint/consistent-type-imports': 'error',
16       '@typescript-eslint/no-misused-promises': ['error', { checksVoidReturn: false }],
17     },
18   },
19   {
20     files: ['**/*.js'],
21     extends: [tseslint.configs.disableTypeChecked],
22   },
23   prettier,
24 );
```

apps/api/.prettierrc.json

JSON · 8 LINES

```
1 {
2   "semi": true,
3   "singleQuote": true,
4   "trailingComma": "all",
5   "printWidth": 100,
6   "tabWidth": 2,
7   "arrowParens": "always"
8 }
```

apps/api/.prettierignore

IGNORE · 4 LINES

```
1 dist
2 node_modules
3 src/generated
4 prisma/migrations
```

apps/api/.dockerignore

SH · 8 LINES

```
1 node_modules
2 dist
3 src/generated
4 .env
5 .env.*
6 !.env.example
7 *.log
8 .DS_Store
```



```

1 # syntax=docker/dockerfile:1
2
3 FROM node:24-alpine AS build
4 WORKDIR /app
5
6 COPY package.json package-lock.json ./
7 RUN npm ci
8
9 COPY prisma.config.ts tsconfig.json tsconfig.build.json ./
10 COPY prisma ./prisma
11 COPY src ./src
12
13 # Client is gitignored, generate it before tsc compiles it into dist.
14 RUN npx prisma generate
15 RUN npm run build
16
17 RUN npm ci --omit=dev
18
19 FROM node:24-alpine AS runtime
20 WORKDIR /app
21 ENV NODE_ENV=production
22
23 # Forwards SIGTERM so `docker compose down` shuts down cleanly.
24 RUN apk add --no-cache dumb-init
25
26 COPY --from=build /app/node_modules ./node_modules
27 COPY --from=build /app/dist ./dist
28 COPY --from=build /app/prisma ./prisma
29 COPY --from=build /app/package.json /app/prisma.config.ts ./
30 COPY docker-entrypoint.sh /usr/local/bin/docker-entrypoint.sh
31 RUN chmod +x /usr/local/bin/docker-entrypoint.sh
32
33 USER node
34 EXPOSE 4000
35
36 ENTRYPOINT ["dumb-init", "--", "/usr/local/bin/docker-entrypoint.sh"]
37 CMD ["node", "dist/server.js"]

```

```

1 #!/bin/sh
2 set -e
3
4 # Generated at build time, not here: the generator emits TypeScript and there is
5 # no src/ in this image to emit into.
6 if [ ! -f dist/generated/prisma/client.js ]; then
7     echo "[entrypoint] FATAL: generated Prisma client missing from the image."
8     echo "[entrypoint] Rebuild with: docker compose build --no-cache api"
9     exit 1
10 fi
11
12 echo "[entrypoint] applying migrations..."
13 ./node_modules/.bin/prisma migrate deploy
14
15 if [ "${SEED_ON_START}" = "true" ]; then
16     echo "[entrypoint] seeding..."
17     node dist/seed.js
18 fi
19
20 echo "[entrypoint] starting API"
21 exec "$@"

```

```

1 import 'dotenv/config';
2 import { defineConfig } from 'prisma/config';
3
4 export default defineConfig({
5   schema: 'prisma/schema.prisma',
6   migrations: {
7     path: 'prisma/migrations',
8   },
9   datasource: {
10     url: process.env['DATABASE_URL'],
11   },
12 });

```

```

1 generator client {
2   provider = "prisma-client"
3   output   = "../src/generated/prisma"
4 }
5
6 datasource db {
7   provider = "postgresql"
8 }
9
10 enum DefectType {
11   WEAVE_DEFECT
12   SHADE_VARIATION
13   HOLE_TEAR
14   COUNT_DEVIATION
15   OTHER
16 }
17
18 enum Severity {
19   CRITICAL
20   MAJOR
21   MINOR
22 }
23
24 enum InspectionStatus {
25   OPEN
26   RESOLVED
27 }
28
29 enum InspectionSource {
30   MANUAL
31   SAP_WEBHOOK
32 }
33
34 model Inspection {
35   id String @id @default(uuid())
36
37   // Plain DATE. only the day matters.
38   inspectionDate DateTime @map("inspection_date") @db.Date
39
40   machineId String @map("machine_id") @db.VarChar(64)
41   defectType DefectType @map("defect_type")
42   severity Severity
43   remarks String? @db.VarChar(2000)
44
45   status InspectionStatus @default(OPEN)
46   resolutionNote String? @map("resolution_note") @db.VarChar(2000)
47   resolvedAt DateTime? @map("resolved_at")
48
49   source InspectionSource @default(MANUAL)

```

```

50
51 // Idempotency key. SAP sets sap:<notification_no>.
52 clientRef String? @unique @map("client_ref") @db.VarChar(64)
53
54 sapPayload Json? @map("sap_payload")
55
56 createdAt DateTime @default(now()) @map("created_at")
57 updatedAt DateTime @updatedAt @map("updated_at")
58
59 @@index([status, severity])
60 @@index([inspectionDate(sort: Desc)])
61 @@index([createdAt(sort: Desc)])
62 @map("inspections")
63 }
64
65 model User {
66   id          String    @id @default(uuid())
67   username    String    @unique @db.VarChar(64)
68   passwordHash String    @map("password_hash")
69   displayName String    @map("display_name") @db.VarChar(120)
70   createdAt   DateTime  @default(now()) @map("created_at")
71
72   @@map("users")
73 }

```

apps/api/prisma/migrations/20260820122703_init/migration.sql

SQL · 57 LINES

```

1  -- CreateEnum
2  CREATE TYPE "DefectType" AS ENUM ('WEAVE_DEFECT', 'SHADE_VARIATION', 'HOLE_TEAR', 'COUNT_DEVIATION',
3  'OTHER');
4
5  -- CreateEnum
6  CREATE TYPE "Severity" AS ENUM ('CRITICAL', 'MAJOR', 'MINOR');
7
8  -- CreateEnum
9  CREATE TYPE "InspectionStatus" AS ENUM ('OPEN', 'RESOLVED');
10
11 -- CreateEnum
12 CREATE TYPE "InspectionSource" AS ENUM ('MANUAL', 'SAP_WEBHOOK');
13
14 -- CreateTable
15 CREATE TABLE "inspections" (
16   "id" TEXT NOT NULL,
17   "inspection_date" DATE NOT NULL,
18   "machine_id" VARCHAR(64) NOT NULL,
19   "defect_type" "DefectType" NOT NULL,
20   "severity" "Severity" NOT NULL,
21   "remarks" VARCHAR(2000),
22   "status" "InspectionStatus" NOT NULL DEFAULT 'OPEN',
23   "resolution_note" VARCHAR(2000),
24   "resolved_at" TIMESTAMP(3),
25   "source" "InspectionSource" NOT NULL DEFAULT 'MANUAL',
26   "client_ref" VARCHAR(64),
27   "sap_payload" JSONB,
28   "created_at" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
29   "updated_at" TIMESTAMP(3) NOT NULL,
30
31   CONSTRAINT "inspections_pkey" PRIMARY KEY ("id")
32 );
33
34 -- CreateTable
35 CREATE TABLE "users" (
36   "id" TEXT NOT NULL,
37   "username" VARCHAR(64) NOT NULL,
38   "password_hash" TEXT NOT NULL,
39   "display_name" VARCHAR(120) NOT NULL,
40   "created_at" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,

```

```

40
41     CONSTRAINT "users_pkey" PRIMARY KEY ("id")
42 );
43
44 -- CreateIndex
45 CREATE UNIQUE INDEX "inspections_client_ref_key" ON "inspections"("client_ref");
46
47 -- CreateIndex
48 CREATE INDEX "inspections_status_severity_idx" ON "inspections"("status", "severity");
49
50 -- CreateIndex
51 CREATE INDEX "inspections_inspection_date_idx" ON "inspections"("inspection_date" DESC);
52
53 -- CreateIndex
54 CREATE INDEX "inspections_created_at_idx" ON "inspections"("created_at" DESC);
55
56 -- CreateIndex
57 CREATE UNIQUE INDEX "users_username_key" ON "users"("username");

```

apps/api/prisma/migrations/migration_lock.toml

TOML · 3 LINES

```

1 # Please do not edit this file manually
2 # It should be added in your version-control system (e.g., Git)
3 provider = "postgresql"

```

apps/api/src/server.ts

TS · 19 LINES

```

1 import { createApp } from './app.ts';
2 import { env } from './env.ts';
3 import { disconnectPrisma } from './prisma.ts';
4
5 const app = createApp();
6
7 const server = app.listen(env.PORT, () => {
8   console.log(`[api] listening on http://localhost:${env.PORT} (${env.NODE_ENV})`);
9 });
10
11 function shutdown(signal: string) {
12   console.log(`[api] ${signal} received, shutting down`);
13   server.close(() => {
14     void disconnectPrisma().finally(() => process.exit(0));
15   });
16 }
17
18 process.on('SIGTERM', () => shutdown('SIGTERM'));
19 process.on('SIGINT', () => shutdown('SIGINT'));

```

apps/api/src/app.ts

TS · 37 LINES

```

1 import express from 'express';
2 import cors from 'cors';
3 import helmet from 'helmet';
4 import morgan from 'morgan';
5 import { env, isProduction } from './env.ts';
6 import { errorHandler, notFoundHandler } from './middleware/errorHandler.ts';
7 import { inspectionRoutes } from './modules/inspections/inspection.routes.ts';
8 import { authRoutes } from './modules/auth/auth.routes.ts';
9 import { sapRoutes } from './modules/sap/sap.routes.ts';
10
11 export function createApp() {
12   const app = express();
13
14   app.disable('x-powered-by');
15   app.set('trust proxy', 1);
16
17   app.use(helmet());

```

```

18 app.use(
19   cors({
20     origin: env.CORS_ORIGIN === '*' ? true : env.CORS_ORIGIN.split(',').map((o) => o.trim()),
21   }),
22 );
23 app.use(express.json({ limit: '256kb' }));
24 app.use(morgan(isProduction ? 'combined' : 'dev'));
25
26 app.get('/api/health', (_req, res) => {
27   res.json({ status: 'ok', uptime: process.uptime(), timestamp: new Date().toISOString() });
28 });
29 app.use('/api/auth', authRoutes);
30 app.use('/api/inspections', inspectionRoutes);
31 app.use('/api/sap-webhook', sapRoutes);
32
33 app.use(notFoundHandler);
34 app.use(errorHandler);
35
36 return app;
37 }

```

apps/api/src/env.ts

TS · 36 LINES

```

1 import 'dotenv/config';
2 import { z } from 'zod';
3
4 const envSchema = z.object({
5   NODE_ENV: z.enum(['development', 'test', 'production']).default('development'),
6   PORT: z.coerce.number().int().positive().default(4000),
7
8   DATABASE_URL: z.string().min(1, 'DATABASE_URL is required'),
9
10  // Comma-separated, or * for any origin.
11  CORS_ORIGIN: z.string().default('*'),
12
13  JWT_SECRET: z
14    .string()
15    .min(16)
16    .default('a9e3feaf15997f1c948d0f240a2382589a59aa98a0a6e262395feff3b11f8d36'),
17  JWT_EXPIRES_IN: z.string().default('12h'),
18
19  // When set, the webhook requires a matching x-sap-signature header.
20  SAP_WEBHOOK_SECRET: z.string().optional(),
21 });
22
23 const parsed = envSchema.safeParse(process.env);
24
25 if (!parsed.success) {
26   const details = parsed.error.issues
27     .map((issue) => ` - ${issue.path.join('.')} || '(root)': ${issue.message}`)
28     .join('\n');
29   console.error(`Invalid environment configuration:\n${details}\n`);
30   process.exit(1);
31 }
32
33 export const env = parsed.data;
34 export type Env = typeof env;
35
36 export const isProduction = env.NODE_ENV === 'production';

```

```

1 import { PrismaPg } from '@prisma/adapters-pg';
2 import { PrismaClient } from './generated/prisma/client.ts';
3 import { env, isProduction } from './env.ts';
4
5 const adapter = new PrismaPg({ connectionString: env.DATABASE_URL });
6
7 export const prisma = new PrismaClient({
8   adapter,
9   log: isProduction ? ['warn', 'error'] : ['warn', 'error'],
10 });
11
12 export async function disconnectPrisma(): Promise<void> {
13   await prisma.$disconnect();
14 }

```

```

1 import bcrypt from 'bcryptjs';
2 import { PrismaPg } from '@prisma/adapters-pg';
3 import { PrismaClient } from './generated/prisma/client.ts';
4 import type { Prisma } from './generated/prisma/client.ts';
5 import 'dotenv/config';
6
7 const connectionString = process.env['DATABASE_URL'];
8 if (!connectionString) {
9   console.error('DATABASE_URL is not set - cannot seed.');
```

```

10   process.exit(1);
11 }
12
13 const prisma = new PrismaClient({ adapter: new PrismaPg({ connectionString }) });
14
15 function daysAgo(n: number): string {
16   const date = new Date();
17   date.setUTCDate(date.getUTCDate() - n);
18   return date.toISOString().slice(0, 10);
19 }
20
21 function dateOnly(value: string): Date {
22   return new Date(`${value}T00:00:00.000Z`);
23 }
24
25 type SeedRow = {
26   days: number;
27   machineId: string;
28   defectType: Prisma.InspectionCreateInput['defectType'];
29   severity: Prisma.InspectionCreateInput['severity'];
30   remarks?: string;
31   resolution?: string;
32 };
33
34 const ROWS: SeedRow[] = [
35   {
36     days: 0,
37     machineId: 'LOOM-14',
38     defectType: 'WEAVE_DEFECT',
39     severity: 'CRITICAL',
40     remarks: 'Broken pick repeating every ~2m across full width.',
41   },
42   {
43     days: 0,
44     machineId: 'LOOM-07',
45     defectType: 'SHADE_VARIATION',
46     severity: 'MAJOR',
47     remarks: 'Selvedge to centre shade drift on indigo lot.',

```

```

48 },
49 {
50   days: 0,
51   machineId: 'SULZER-03',
52   defectType: 'COUNT_DEVIATION',
53   severity: 'MINOR',
54   remarks: 'EPI reading 118 against standard 124.',
55 },
56 {
57   days: 1,
58   machineId: 'LOOM-14',
59   defectType: 'HOLE_TEAR',
60   severity: 'CRITICAL',
61   remarks: 'Temple mark tearing at right selvage.',
62 },
63 { days: 1, machineId: 'AIRJET-21', defectType: 'WEAVE_DEFECT', severity: 'MAJOR' },
64 {
65   days: 2,
66   machineId: 'LOOM-09',
67   defectType: 'OTHER',
68   severity: 'MINOR',
69   remarks: 'Oil spot near beam, suspect lubrication overflow.',
70   resolution: 'Cleaned beam housing and wiped down guide rollers. Next 200m clear.',
71 },
72 {
73   days: 3,
74   machineId: 'SULZER-03',
75   defectType: 'SHADE_VARIATION',
76   severity: 'CRITICAL',
77   remarks: 'Batch-to-batch mismatch flagged by finishing.',
78   resolution: 'Lot quarantined and re-dyed. Dyehouse recipe corrected.',
79 },
80 { days: 3, machineId: 'LOOM-07', defectType: 'WEAVE_DEFECT', severity: 'MINOR' },
81 {
82   days: 4,
83   machineId: 'RAPIER-11',
84   defectType: 'COUNT_DEVIATION',
85   severity: 'MAJOR',
86   remarks: 'PPI short by 4 on three consecutive rolls.',
87   resolution: 'Let-off tension recalibrated by maintenance. Verified on sample.',
88 },
89 {
90   days: 5,
91   machineId: 'LOOM-14',
92   defectType: 'HOLE_TEAR',
93   severity: 'MINOR',
94   resolution: 'Isolated pinhole, roll downgraded to B-grade and passed.',
95 },
96 {
97   days: 6,
98   machineId: 'AIRJET-21',
99   defectType: 'SHADE_VARIATION',
100   severity: 'MAJOR',
101   remarks: 'Visible barre under D65 light box.',
102 },
103 {
104   days: 7,
105   machineId: 'LOOM-02',
106   defectType: 'WEAVE_DEFECT',
107   severity: 'CRITICAL',
108   remarks: 'Reed mark running full roll length.',
109   resolution: 'Reed replaced. Loom re-qualified with a 50m trial run.',
110 },
111 {
112   days: 8,
113   machineId: 'SULZER-03',
114   defectType: 'OTHER',
115   severity: 'MAJOR',

```

```

116     remarks: 'Unidentified contamination streak, sample sent to lab.',
117 },
118 {
119     days: 9,
120     machineId: 'RAPIER-11',
121     defectType: 'HOLE_TEAR',
122     severity: 'CRITICAL',
123     resolution: 'Gripper tape frayed; replaced and stitched section cut out.',
124 },
125 { days: 11, machineId: 'LOOM-09', defectType: 'COUNT_DEVIATION', severity: 'MINOR' },
126 {
127     days: 12,
128     machineId: 'LOOM-07',
129     defectType: 'HOLE_TEAR',
130     severity: 'MAJOR',
131     remarks: 'Two tears within 15m near left selvedge.',
132 },
133 {
134     days: 14,
135     machineId: 'AIRJET-21',
136     defectType: 'COUNT_DEVIATION',
137     severity: 'CRITICAL',
138     remarks: 'Count off by 9 EPI – whole shift output held.',
139     resolution: 'Warp beam mis-set at creel. Shift output re-inspected, 60% released.',
140 },
141 {
142     days: 16,
143     machineId: 'LOOM-02',
144     defectType: 'SHADE_VARIATION',
145     severity: 'MINOR',
146     resolution: 'Within tolerance on re-check. Closed with no action.',
147 },
148 { days: 18, machineId: 'SULZER-08', defectType: 'WEAVE_DEFECT', severity: 'MAJOR' },
149 {
150     days: 21,
151     machineId: 'LOOM-14',
152     defectType: 'OTHER',
153     severity: 'MINOR',
154     remarks: 'Roll label mismatch against production docket.',
155     resolution: 'Docket corrected at packing. Root cause: manual entry.',
156 },
157 {
158     days: 24,
159     machineId: 'RAPIER-11',
160     defectType: 'WEAVE_DEFECT',
161     severity: 'MINOR',
162     resolution: 'Minor float, trimmed at inspection table.',
163 },
164 {
165     days: 28,
166     machineId: 'SULZER-08',
167     defectType: 'HOLE_TEAR',
168     severity: 'MAJOR',
169     remarks: 'Recurring at same loom position.',
170 },
171 {
172     days: 31,
173     machineId: 'LOOM-09',
174     defectType: 'SHADE_VARIATION',
175     severity: 'CRITICAL',
176     resolution: 'Full lot re-processed. Supplier yarn batch blacklisted.',
177 },
178 {
179     days: 35,
180     machineId: 'LOOM-02',
181     defectType: 'COUNT_DEVIATION',
182     severity: 'MAJOR',
183     resolution: 'Tension roller serviced during planned maintenance.',

```



```

184 },
185 {
186   days: 40,
187   machineId: 'AIRJET-21',
188   defectType: 'OTHER',
189   severity: 'MINOR',
190   remarks: 'Historical entry migrated from paper register.',
191   resolution: 'Closed during backlog cleanup, no action required.',
192 },
193 ];
194
195 async function main() {
196   // Entrypoint runs this on every boot, so don't wipe unless asked.
197   const force = process.env['SEED_FORCE'] === 'true';
198   const existing = await prisma.inspection.count();
199
200   if (existing > 0 && !force) {
201     console.log(`[seed] skipped - ${existing} inspections already present`);
202     console.log('[seed] set SEED_FORCE=true to wipe and reseed');
203     return;
204   }
205
206   console.log('[seed] clearing existing data...');
207   await prisma.inspection.deleteMany();
208   await prisma.user.deleteMany();
209
210   const passwordHash = await bcrypt.hash('arvind123', 10);
211   await prisma.user.create({
212     data: { username: 'supervisor', passwordHash, displayName: 'Shop-floor Supervisor' },
213   });
214   console.log('[seed] user created: supervisor / arvind123');
215
216   for (const row of ROWS) {
217     const resolved = Boolean(row.resolution);
218     await prisma.inspection.create({
219       data: {
220         inspectionDate: dateOnly(daysAgo(row.days)),
221         machineId: row.machineId,
222         defectType: row.defectType,
223         severity: row.severity,
224         remarks: row.remarks ?? null,
225         status: resolved ? 'RESOLVED' : 'OPEN',
226         resolutionNote: row.resolution ?? null,
227         resolvedAt: resolved ? new Date(dateOnly(daysAgo(Math.max(0, row.days - 1)))) : null,
228       },
229     });
230   }
231
232   const open = await prisma.inspection.count({ where: { status: 'OPEN' } });
233   const resolved = await prisma.inspection.count({ where: { status: 'RESOLVED' } });
234   console.log(`[seed] ${ROWS.length} inspections created (${open} open, ${resolved} resolved)`);
235 }
236
237 main()
238   .catch((error: unknown) => {
239     console.error('[seed] failed:', error);
240     process.exitCode = 1;
241   })
242   .finally(() => {
243     void prisma.$disconnect();
244   });

```

```

1 import type { NextFunction, Request, RequestHandler, Response } from 'express';
2
3 export const asyncHandler =
4   <T extends RequestHandler>(handler: T): RequestHandler =>
5     (req: Request, res: Response, next: NextFunction) => {
6       void Promise.resolve(handler(req, res, next)).catch(next);
7     };

```

```

1 import { badRequest } from './errors.ts';
2
3 const DATE_ONLY = /^d{4}-d{2}-d{2}$/;
4
5 // Pinned to UTC midnight. Local-time Dates shift the day across the boundary.
6 export function parseDateOnly(value: string, field = 'date'): Date {
7   if (!DATE_ONLY.test(value)) {
8     throw badRequest(`${field} must be formatted as YYYY-MM-DD`);
9   }
10  const date = new Date(`${value}T00:00:00.000Z`);
11  if (Number.isNaN(date.getTime())) {
12    throw badRequest(`${field} is not a valid calendar date`);
13  }
14  return date;
15 }
16
17 export function formatDateOnly(date: Date): string {
18   return date.toISOString().slice(0, 10);
19 }
20
21 export function endOfDayExclusive(value: string, field = 'date'): Date {
22   const date = parseDateOnly(value, field);
23   date.setUTCDate(date.getUTCDate() + 1);
24   return date;
25 }

```

```

1 export class AppError extends Error {
2   readonly statusCode: number;
3   readonly code: string;
4   readonly details?: unknown;
5
6   constructor(statusCode: number, code: string, message: string, details?: unknown) {
7     super(message);
8     this.name = 'AppError';
9     this.statusCode = statusCode;
10    this.code = code;
11    this.details = details;
12  }
13 }
14
15 export const badRequest = (message: string, details?: unknown) =>
16   new AppError(400, 'BAD_REQUEST', message, details);
17
18 export const unauthorized = (message = 'Authentication required') =>
19   new AppError(401, 'UNAUTHORIZED', message);
20
21 export const notFound = (message = 'Resource not found') => new AppError(404, 'NOT_FOUND', message);
22
23 export const conflict = (message: string, details?: unknown) =>
24   new AppError(409, 'CONFLICT', message, details);

```

```

1  import type { RequestHandler } from 'express';
2  import jwt from 'jsonwebtoken';
3  import { env } from '../env.ts';
4  import { unauthorized } from '../lib/errors.ts';
5
6  export interface AuthTokenPayload {
7    sub: string;
8    username: string;
9    displayName: string;
10 }
11
12 declare global {
13   // eslint-disable-next-line @typescript-eslint/no-namespace
14   namespace Express {
15     interface Request {
16       user?: AuthTokenPayload;
17     }
18   }
19 }
20
21 export function signToken(payload: AuthTokenPayload): string {
22   return jwt.sign(payload, env.JWT_SECRET, {
23     expiresIn: env.JWT_EXPIRES_IN as jwt.SignOptions['expiresIn'],
24   });
25 }
26
27 function readBearer(header: string | undefined): string | null {
28   if (!header) return null;
29   const [scheme, token] = header.split(' ');
30   if (!token || scheme?.toLowerCase() !== 'bearer') return null;
31   return token;
32 }
33
34 export const requireAuth: RequestHandler = (req, _res, next) => {
35   const token = readBearer(req.headers.authorization);
36   if (!token) {
37     next(unauthorized('Missing bearer token'));
38     return;
39   }
40
41   try {
42     req.user = jwt.verify(token, env.JWT_SECRET) as AuthTokenPayload;
43     next();
44   } catch {
45     next(unauthorized('Token is invalid or has expired'));
46   }
47 };

```

```

1  import type { ErrorRequestHandler, RequestHandler } from 'express';
2  import { ZodError } from 'zod';
3  import { Prisma } from '../generated/prisma/client.ts';
4  import { AppError } from '../lib/errors.ts';
5  import { isProduction } from '../env.ts';
6
7  export interface ApiErrorBody {
8    error: {
9      code: string;
10     message: string;
11     details?: unknown;
12   };
13 }
14

```

```

15 // { field: [messages] } so the client can highlight inputs.
16 function fieldErrors(error: ZodError): Record<string, string[]> {
17   const result: Record<string, string[]> = {};
18   for (const issue of error.issues) {
19     const key = issue.path.join('.') || '_';
20     (result[key] ?? []).push(issue.message);
21   }
22   return result;
23 }
24
25 // Environment problems, not caller mistakes. Point at the missed setup step.
26 const DATABASE_ERROR_MESSAGES: Record<string, string> = {
27   ECONNREFUSED:
28     'Cannot reach PostgreSQL. Is the database running on the host/port in DATABASE_URL?',
29   ENOTFOUND: 'The database host in DATABASE_URL could not be resolved.',
30   EAI_AGAIN: 'The database host in DATABASE_URL could not be resolved.',
31   ETIMEDOUT: 'Timed out connecting to PostgreSQL.',
32   ECONNRESET: 'The database connection was reset.',
33   P1001: 'Cannot reach PostgreSQL. Is the database running on the host/port in DATABASE_URL?',
34   P1002: 'Timed out connecting to PostgreSQL.',
35   P1017: 'PostgreSQL closed the connection.',
36   P2024: 'Timed out acquiring a database connection from the pool.',
37   '28P01': 'PostgreSQL rejected the credentials in DATABASE_URL.',
38   '3D000': 'The database named in DATABASE_URL does not exist.',
39   '42P01': 'Tables are missing. Run `npm run db:deploy` (or `db:migrate`) to apply migrations.',
40 };
41
42 export const notFoundHandler: RequestHandler = (req, res) => {
43   const body: ApiErrorBody = {
44     error: { code: 'NOT_FOUND', message: `No route matches ${req.method} ${req.originalUrl}` },
45   };
46   res.status(404).json(body);
47 };
48
49 export const errorHandler: ErrorRequestHandler = (err, _req, res, _next) => {
50   if (err instanceof AppError) {
51     const body: ApiErrorBody = {
52       error: { code: err.code, message: err.message, details: err.details },
53     };
54     res.status(err.statusCode).json(body);
55     return;
56   }
57
58   if (err instanceof ZodError) {
59     const body: ApiErrorBody = {
60       error: {
61         code: 'VALIDATION_ERROR',
62         message: 'One or more fields are invalid',
63         details: fieldErrors(err),
64       },
65     };
66     res.status(422).json(body);
67     return;
68   }
69
70   if (err instanceof Prisma.PrismaClientKnownRequestError) {
71     if (err.code === 'P2002') {
72       res.status(409).json({
73         error: { code: 'DUPLICATE', message: 'A record with this reference already exists' },
74       } satisfies ApiErrorBody);
75       return;
76     }
77     if (err.code === 'P2025') {
78       res.status(404).json({
79         error: { code: 'NOT_FOUND', message: 'Resource not found' },
80       } satisfies ApiErrorBody);
81       return;
82     }

```

```

83 }
84
85 const bodyParserType = (err as { type?: string }).type;
86 if (bodyParserType === 'entity.parse.failed') {
87   res.status(400).json({
88     error: { code: 'MALFORMED_JSON', message: 'Request body is not valid JSON' },
89   } satisfies ApiErrorBody);
90   return;
91 }
92 if (bodyParserType === 'entity.too.large') {
93   res.status(413).json({
94     error: { code: 'PAYLOAD_TOO_LARGE', message: 'Request body exceeds the 256kb limit' },
95   } satisfies ApiErrorBody);
96   return;
97 }
98
99 const infraMessage = DATABASE_ERROR_MESSAGES[String((err as { code?: string }).code)];
100 if (infraMessage || err instanceof Prisma.PrismaClientInitializationError) {
101   console.error('[db] unavailable:', (err as Error).message);
102   res.status(503).json({
103     error: {
104       code: 'DATABASE_UNAVAILABLE',
105       message: infraMessage ?? 'Cannot reach the database.',
106     },
107   } satisfies ApiErrorBody);
108   return;
109 }
110
111 console.error('[api] unhandled error:', err);
112 res.status(500).json({
113   error: {
114     code: 'INTERNAL_ERROR',
115     message: 'Something went wrong',
116     details: isProduction ? undefined : String(err instanceof Error ? err.stack : err),
117   },
118 } satisfies ApiErrorBody);
119 };

```

apps/api/src/modules/auth/auth.routes.ts

TS · 36 LINES

```

1 import { Router } from 'express';
2 import bcrypt from 'bcryptjs';
3 import { z } from 'zod';
4 import { prisma } from '../../prisma.ts';
5 import { asyncHandler } from '../../lib/asyncHandler.ts';
6 import { unauthorized } from '../../lib/errors.ts';
7 import { requireAuth, signToken } from '../../middleware/auth.ts';
8
9 const loginSchema = z.object({
10   username: z.string().trim().min(1, 'Username is required'),
11   password: z.string().min(1, 'Password is required'),
12 });
13
14 export const authRoutes = Router();
15
16 authRoutes.post(
17   '/login',
18   asyncHandler(async (req, res) => {
19     const { username, password } = loginSchema.parse(req.body);
20
21     const user = await prisma.user.findUnique({
22       where: { username: username.toLowerCase() },
23     });
24
25     if (!user || !(await bcrypt.compare(password, user.passwordHash))) {
26       throw unauthorized('Incorrect username or password');
27     }

```

```

28
29     const payload = { sub: user.id, username: user.username, displayName: user.displayName };
30     res.json({ token: signToken(payload), user: payload });
31   }),
32 );
33
34 authRoutes.get('/me', requireAuth, (req, res) => {
35   res.json({ user: req.user ?? null });
36 });

```

apps/api/src/modules/inspections/inspection.mapper.ts

TS · 36 LINES

```

1  import type { Inspection } from '../../generated/prisma/client.ts';
2  import { formatDateOnly } from '../../lib/dates.ts';
3
4  export interface InspectionDTO {
5    id: string;
6    inspectionDate: string;
7    machineId: string;
8    defectType: Inspection['defectType'];
9    severity: Inspection['severity'];
10   remarks: string | null;
11   status: Inspection['status'];
12   resolutionNote: string | null;
13   resolvedAt: string | null;
14   source: Inspection['source'];
15   clientRef: string | null;
16   createdAt: string;
17   updatedAt: string;
18 }
19
20 export function toInspectionDTO(row: Inspection): InspectionDTO {
21   return {
22     id: row.id,
23     inspectionDate: formatDateOnly(row.inspectionDate),
24     machineId: row.machineId,
25     defectType: row.defectType,
26     severity: row.severity,
27     remarks: row.remarks,
28     status: row.status,
29     resolutionNote: row.resolutionNote,
30     resolvedAt: row.resolvedAt ? row.resolvedAt.toISOString() : null,
31     source: row.source,
32     clientRef: row.clientRef,
33     createdAt: row.createdAt.toISOString(),
34     updatedAt: row.updatedAt.toISOString(),
35   };
36 }

```

apps/api/src/modules/inspections/inspection.routes.ts

TS · 58 LINES

```

1  import { Router } from 'express';
2  import { asyncHandler } from '../../lib/asyncHandler.ts';
3  import { requireAuth } from '../../middleware/auth.ts';
4  import {
5    createInspectionSchema,
6    listInspectionsSchema,
7    resolveInspectionSchema,
8  } from './inspection.schemas.ts';
9  import {
10    createInspection,
11    getInspection,
12    getSummary,
13    listInspections,
14    resolveInspection,
15  } from './inspection.service.ts';

```

```

16
17 export const inspectionRoutes = Router();
18
19 inspectionRoutes.use(requireAuth);
20
21 inspectionRoutes.get(
22   '/summary',
23   asyncHandler(async (_req, res) => {
24     res.json(await getSummary());
25   }),
26 );
27
28 inspectionRoutes.get(
29   '/',
30   asyncHandler(async (req, res) => {
31     const query = listInspectionsSchema.parse(req.query);
32     res.json(await listInspections(query));
33   }),
34 );
35
36 inspectionRoutes.get(
37   '/:id',
38   asyncHandler(async (req, res) => {
39     res.json(await getInspection(String(req.params.id)));
40   }),
41 );
42
43 inspectionRoutes.post(
44   '/',
45   asyncHandler(async (req, res) => {
46     const input = createInspectionSchema.parse(req.body);
47     const { inspection } = await createInspection(input);
48     res.status(201).json(inspection);
49   }),
50 );
51
52 inspectionRoutes.patch(
53   '/:id/resolve',
54   asyncHandler(async (req, res) => {
55     const input = resolveInspectionSchema.parse(req.body);
56     res.json(await resolveInspection(String(req.params.id), input));
57   }),
58 );

```

apps/api/src/modules/inspections/inspection.schemas.ts

TS · 67 LINES

```

1 import { z } from 'zod';
2
3 export const defectTypeSchema = z.enum([
4   'WEAVE_DEFECT',
5   'SHADE_VARIATION',
6   'HOLE_TEAR',
7   'COUNT_DEVIATION',
8   'OTHER',
9 ]);
10
11 export const severitySchema = z.enum(['CRITICAL', 'MAJOR', 'MINOR']);
12 export const statusSchema = z.enum(['OPEN', 'RESOLVED']);
13 export const sourceSchema = z.enum(['MANUAL', 'SAP_WEBHOOK']);
14
15 const dateOnlySchema = z.string().regex(/^\d{4}-\d{2}-\d{2}$/, 'Use the YYYY-MM-DD format');
16
17 const optionalText = (max: number) =>
18   z
19     .string()
20     .trim()
21     .max(max, `Must be ${max} characters or fewer`)

```

```

22     .nullish()
23     .transform((value) => (value && value.length > 0 ? value : null));
24
25 export const createInspectionSchema = z.object({
26   inspectionDate: dateOnlySchema,
27   machineId: z
28     .string()
29     .trim()
30     .min(1, 'Machine / line ID is required')
31     .max(64, 'Must be 64 characters or fewer'),
32   defectType: defectTypeSchema,
33   severity: severitySchema,
34   remarks: optionalText(2000),
35 });
36
37 export const resolveInspectionSchema = z.object({
38   resolutionNote: z
39     .string()
40     .trim()
41     .min(1, 'A resolution note is required to close an inspection')
42     .max(2000, 'Must be 2000 characters or fewer'),
43 });
44
45 export const sortBySchema = z.enum(['inspectionDate', 'createdAt', 'severity', 'machineId']);
46
47 export const listInspectionsSchema = z
48   .object({
49     status: statusSchema.optional(),
50     severity: severitySchema.optional(),
51     defectType: defectTypeSchema.optional(),
52     machineId: z.string().trim().min(1).optional(),
53     from: dateOnlySchema.optional(),
54     to: dateOnlySchema.optional(),
55     sortBy: sortBySchema.default('inspectionDate'),
56     order: z.enum(['asc', 'desc']).default('desc'),
57     page: z.coerce.number().int().min(1).default(1),
58     limit: z.coerce.number().int().min(1).max(100).default(20),
59   })
60   .refine((value) => !value.from || !value.to || value.from <= value.to, {
61     message: '"from" must be on or before "to"',
62     path: ['from'],
63   });
64
65 export type CreateInspectionInput = z.infer<typeof createInspectionSchema>;
66 export type ResolveInspectionInput = z.infer<typeof resolveInspectionSchema>;
67 export type ListInspectionsQuery = z.infer<typeof listInspectionsSchema>;

```

apps/api/src/modules/inspections/inspection.service.ts

TS · 182 LINES

```

1 import type { Prisma } from '../../generated/prisma/client.ts';
2 import { prisma } from '../../prisma.ts';
3 import { conflict, notFound } from '../../lib/errors.ts';
4 import { endOfDayExclusive, parseDateOnly } from '../../lib/dates.ts';
5 import { toInspectionDTO, type InspectionDTO } from './inspection.mapper.ts';
6 import type {
7   CreateInspectionInput,
8   ListInspectionsQuery,
9   ResolveInspectionInput,
10 } from './inspection.schemas.ts';
11
12 export function normaliseMachineId(raw: string): string {
13   return raw.trim().replace(/\s+/g, ' ').toUpperCase();
14 }
15
16 export interface CreateOptions {
17   source?: 'MANUAL' | 'SAP_WEBHOOK';
18   sapPayload?: Prisma.InputJsonValue;

```



```

19   clientRef?: string | null;
20 }
21
22 export interface CreateResult {
23   inspection: InspectionDTO;
24   created: boolean;
25 }
26
27 export async function createInspection(
28   input: CreateInspectionInput,
29   options: CreateOptions = {},
30 ): Promise<CreateResult> {
31   const clientRef = options.clientRef ?? null;
32
33   // Replayed webhook delivery, return what we already have.
34   if (clientRef) {
35     const existing = await prisma.inspection.findUnique({ where: { clientRef } });
36     if (existing) {
37       return { inspection: toInspectionDTO(existing), created: false };
38     }
39   }
40
41   const data: Prisma.InspectionCreateInput = {
42     inspectionDate: parseDateOnly(input.inspectionDate, 'inspectionDate'),
43     machineId: normaliseMachineId(input.machineId),
44     defectType: input.defectType,
45     severity: input.severity,
46     remarks: input.remarks ?? null,
47     source: options.source ?? 'MANUAL',
48     clientRef,
49     ...(options.sapPayload ? { sapPayload: options.sapPayload } : {}),
50   };
51
52   const row = await prisma.inspection.create({ data });
53   return { inspection: toInspectionDTO(row), created: true };
54 }
55
56 export interface ListResult {
57   data: InspectionDTO[];
58   pagination: { page: number; limit: number; total: number; totalPages: number };
59 }
60
61 export async function listInspections(query: ListInspectionsQuery): Promise<ListResult> {
62   const where = buildWhere(query);
63
64   // Enum is declared CRITICAL, MAJOR, MINOR so asc sorts by urgency.
65   const orderBy: Prisma.InspectionOrderByWithRelationInput[] = [{ [query.sortBy]: query.order }];
66   if (query.sortBy !== 'createdAt') {
67     orderBy.push({ createdAt: 'desc' });
68   }
69
70   const [total, rows] = await Promise.all([
71     prisma.inspection.count({ where }),
72     prisma.inspection.findMany({
73       where,
74       orderBy,
75       skip: (query.page - 1) * query.limit,
76       take: query.limit,
77     }),
78   ]);
79
80   return {
81     data: rows.map(toInspectionDTO),
82     pagination: {
83       page: query.page,
84       limit: query.limit,
85       total,
86       totalPages: Math.max(1, Math.ceil(total / query.limit)),

```

```

87     },
88   };
89 }
90
91 function buildWhere(query: ListInspectionsQuery): Prisma.InspectionWhereInput {
92   const where: Prisma.InspectionWhereInput = {};
93
94   if (query.status) where.status = query.status;
95   if (query.severity) where.severity = query.severity;
96   if (query.defectType) where.defectType = query.defectType;
97   if (query.machineId) {
98     where.machineId = { contains: normaliseMachineId(query.machineId) };
99   }
100
101   // Filters on when the defect happened, not when it was entered.
102   if (query.from || query.to) {
103     where.inspectionDate = {
104       ...(query.from ? { gte: parseDateOnly(query.from, 'from') } : {}),
105       ...(query.to ? { lt: endOfDayExclusive(query.to, 'to') } : {}),
106     };
107   }
108
109   return where;
110 }
111
112 export async function getInspection(id: string): Promise<InspectionDTO> {
113   const row = await prisma.inspection.findUnique({ where: { id } });
114   if (!row) throw notFound(`No inspection with id ${id}`);
115   return toInspectionDTO(row);
116 }
117
118 export async function resolveInspection(
119   id: string,
120   input: ResolveInspectionInput,
121 ): Promise<InspectionDTO> {
122   const existing = await prisma.inspection.findUnique({ where: { id } });
123   if (!existing) throw notFound(`No inspection with id ${id}`);
124
125   if (existing.status === 'RESOLVED') {
126     throw conflict('This inspection has already been resolved', {
127       resolvedAt: existing.resolvedAt?.toISOString() ?? null,
128     });
129   }
130
131   const row = await prisma.inspection.update({
132     where: { id },
133     data: {
134       status: 'RESOLVED',
135       resolutionNote: input.resolutionNote,
136       resolvedAt: new Date(),
137     },
138   });
139
140   return toInspectionDTO(row);
141 }
142
143 export interface SeveritySummaryRow {
144   severity: 'CRITICAL' | 'MAJOR' | 'MINOR';
145   open: number;
146   resolved: number;
147   total: number;
148 }
149
150 export interface SummaryResult {
151   totals: { open: number; resolved: number; total: number };
152   bySeverity: SeveritySummaryRow[];
153 }
154

```

```

155 const SEVERITY_ORDER = ['CRITICAL', 'MAJOR', 'MINOR'] as const;
156
157 // Plant-wide on purpose, the list view's filters don't apply here.
158 export async function getSummary(): Promise<SummaryResult> {
159     const grouped = await prisma.inspection.groupBy({
160         by: ['severity', 'status'],
161         _count: { _all: true },
162     });
163
164     const bySeverity: SeveritySummaryRow[] = SEVERITY_ORDER.map((severity) => {
165         const open =
166             grouped.find((g) => g.severity === severity && g.status === 'OPEN')?._count._all ?? 0;
167         const resolved =
168             grouped.find((g) => g.severity === severity && g.status === 'RESOLVED')?._count._all ?? 0;
169         return { severity, open, resolved, total: open + resolved };
170     });
171
172     const totals = bySeverity.reduce(
173         (acc, row) => ({
174             open: acc.open + row.open,
175             resolved: acc.resolved + row.resolved,
176             total: acc.total + row.total,
177         }),
178         { open: 0, resolved: 0, total: 0 },
179     );
180
181     return { totals, bySeverity };
182 }

```

apps/api/src/modules/sap/sap.routes.ts

TS · 92 LINES

```

1 import { Router } from 'express';
2 import { z } from 'zod';
3 import { env } from '../../../env.ts';
4 import { asyncHandler } from '../../../lib/asyncHandler.ts';
5 import { unauthorized } from '../../../lib/errors.ts';
6 import { createInspection } from '../../../inspections/inspection.service.ts';
7 import type { Prisma } from '../../../generated/prisma/client.ts';
8
9 // Mock SAP QM notification. Unknown codes fall back to OTHER/MAJOR rather than
10 // rejecting, otherwise SAP just retries the same bad message forever.
11 const sapPayloadSchema = z.object({
12     notification_no: z.string().trim().min(1).max(48),
13     plant_code: z.string().trim().max(16).optional(),
14     posting_date: z
15         .string()
16         .regex(/^\d{4}-\d{2}-\d{2}$/, 'posting_date must be YYYY-MM-DD')
17         .optional(),
18     work_center: z.string().trim().min(1).max(64),
19     defect_code: z.string().trim().min(1).max(32),
20     // SAP priority scale: 1 = Critical, 2 = Major, 3 = Minor.
21     priority: z.union([z.literal(1), z.literal(2), z.literal(3), z.enum(['1', '2', '3'])]).optional(),
22     long_text: z.string().trim().max(2000).optional(),
23 });
24
25 export type SapPayload = z.infer<typeof sapPayloadSchema>;
26
27 const DEFECT_CODE_MAP: Record<
28     string,
29     'WEAVE_DEFECT' | 'SHADE_VARIATION' | 'HOLE_TEAR' | 'COUNT_DEVIATION'
30 > = {
31     WEAVE: 'WEAVE_DEFECT',
32     WEAVE_DEFECT: 'WEAVE_DEFECT',
33     SHADE: 'SHADE_VARIATION',
34     SHADE_VARIATION: 'SHADE_VARIATION',
35     HOLE: 'HOLE_TEAR',
36     TEAR: 'HOLE_TEAR',

```

```

37 HOLE_TEAR: 'HOLE_TEAR',
38 COUNT: 'COUNT_DEVIATION',
39 COUNT_DEVIATION: 'COUNT_DEVIATION',
40 };
41
42 const PRIORITY_MAP = { '1': 'CRITICAL', '2': 'MAJOR', '3': 'MINOR' } as const;
43
44 function today(): string {
45     return new Date().toISOString().slice(0, 10);
46 }
47
48 export const sapRoutes = Router();
49
50 sapRoutes.post(
51     '/',
52     asyncHandler(async (req, res) => {
53         // Unset in dev so the endpoint stays curl-able.
54         if (env.SAP_WEBHOOK_SECRET && req.headers['x-sap-signature'] !== env.SAP_WEBHOOK_SECRET) {
55             throw unauthorized('Invalid or missing x-sap-signature header');
56         }
57
58         const payload = sapPayloadSchema.parse(req.body);
59
60         const defectType = DEFECT_CODE_MAP[payload.defect_code.toUpperCase()] ?? 'OTHER';
61         const severity = payload.priority
62             ? PRIORITY_MAP[String(payload.priority) as '1' | '2' | '3']
63             : 'MAJOR';
64
65         const remarkParts = [
66             payload.long_text,
67             payload.plant_code ? `Plant ${payload.plant_code}` : null,
68         ].filter((part): part is string => Boolean(part));
69
70         const { inspection, created } = await createInspection(
71             {
72                 inspectionDate: payload.posting_date ?? today(),
73                 machineId: payload.work_center,
74                 defectType,
75                 severity,
76                 remarks: remarkParts.length > 0 ? remarkParts.join(' · ') : null,
77             },
78             {
79                 source: 'SAP_WEBHOOK',
80                 sapPayload: payload as unknown as Prisma.InputJsonValue,
81                 clientRef: `sap:${payload.notification_no}`,
82             },
83         );
84
85         res.status(created ? 201 : 200).json({
86             received: true,
87             created,
88             inspectionId: inspection.id,
89             inspection,
90         });
91     }),
92 );

```

4 · Frontend — apps/web

React 19 + Vite single-page app, served by nginx in production.

apps/web/.env.example

ENV · 3 LINES

```
1 # API Base URL
2 API_PROXY_TARGET=http://localhost:4000
```

apps/web/.gitignore

IGNORE · 24 LINES

```
1 # Logs
2 logs
3 *.log
4 npm-debug.log*
5 yarn-debug.log*
6 yarn-error.log*
7 pnpm-debug.log*
8 lerna-debug.log*
9
10 node_modules
11 dist
12 dist-ssr
13 *.local
14
15 # Editor directories and files
16 .vscode/*
17 !.vscode/extensions.json
18 .idea
19 .DS_Store
20 *.suo
21 *.ntvs*
22 *.njsproj
23 *.sln
24 *.sw?
```

apps/web/package.json

JSON · 42 LINES

```
1 {
2   "name": "web",
3   "private": true,
4   "version": "0.0.0",
5   "type": "module",
6   "scripts": {
7     "dev": "vite",
8     "build": "tsc -b && vite build",
9     "lint": "eslint .",
10    "preview": "vite preview",
11    "lint:fix": "eslint . --fix",
12    "format": "prettier --write \"src/**/*.ts,tsx,css\"",
13    "typecheck": "tsc -b"
14  },
15  "dependencies": {
16    "@hookform/resolvers": "^5.9.1",
17    "@tanstack/react-query": "^5.101.4",
18    "react": "^19.2.8",
19    "react-dom": "^19.2.8",
20    "react-hook-form": "^7.85.0",
21    "react-router": "^8.3.0",
22    "zod": "^4.4.3"
23  },
24  "devDependencies": {
25    "@eslint/js": "^10.0.1",
26    "@tailwindcss/vite": "^4.3.3",
```

```

27   "@types/node": "^24.13.3",
28   "@types/react": "^19.2.17",
29   "@types/react-dom": "^19.2.3",
30   "@vitejs/plugin-react": "^6.0.4",
31   "eslint": "^10.8.1",
32   "eslint-config-prettier": "^10.1.8",
33   "eslint-plugin-react-hooks": "^7.1.1",
34   "eslint-plugin-react-refresh": "^0.5.4",
35   "globals": "^17.11.0",
36   "prettier": "^3.9.6",
37   "tailwindcss": "^4.3.3",
38   "typescript": "~6.0.2",
39   "typescript-eslint": "^8.67.0",
40   "vite": "^8.2.0"
41 }
42 }

```

apps/web/tsconfig.json

JSON · 11 LINES

```

1 {
2   "files": [],
3   "references": [
4     {
5       "path": "./tsconfig.app.json"
6     },
7     {
8       "path": "./tsconfig.node.json"
9     }
10  ]
11 }

```

apps/web/tsconfig.app.json

JSON · 31 LINES

```

1 {
2   "compilerOptions": {
3     "tsBuildInfoFile": "./node_modules/.tmp/tsconfig.app.tsbuildinfo",
4     "target": "es2023",
5     "lib": [
6       "ES2023",
7       "DOM"
8     ],
9     "module": "esnext",
10    "types": [
11      "vite/client"
12    ],
13    "allowArbitraryExtensions": true,
14    "skipLibCheck": true,
15    "moduleResolution": "bundler",
16    "allowImportingTsExtensions": true,
17    "verbatimModuleSyntax": true,
18    "moduleDetection": "force",
19    "noEmit": true,
20    "jsx": "react-jsx",
21    "strict": true,
22    "noUncheckedIndexedAccess": true,
23    "noUnusedLocals": true,
24    "noUnusedParameters": true,
25    "erasableSyntaxOnly": true,
26    "noFallthroughCasesInSwitch": true
27  },
28  "include": [
29    "src"
30  ]
31 }

```

```

1 {
2   "compilerOptions": {
3     "tsBuildInfoFile": "./node_modules/.tmp/tsconfig.node.tsbuildinfo",
4     "target": "es2023",
5     "lib": [
6       "ES2023"
7     ],
8     "types": [
9       "node"
10    ],
11    "skipLibCheck": true,
12    "module": "nodenext",
13    "allowImportingTsExtensions": true,
14    "verbatimModuleSyntax": true,
15    "moduleDetection": "force",
16    "noEmit": true,
17    "noUnusedLocals": true,
18    "noUnusedParameters": true,
19    "erasableSyntaxOnly": true,
20    "noFallthroughCasesInSwitch": true
21  },
22  "include": [
23    "vite.config.ts"
24  ]
25 }

```

```

1 import js from '@eslint/js';
2 import globals from 'globals';
3 import tseslint from 'typescript-eslint';
4 import reactHooks from 'eslint-plugin-react-hooks';
5 import reactRefresh from 'eslint-plugin-react-refresh';
6 import prettier from 'eslint-config-prettier';
7
8 export default tseslint.config(
9   { ignores: ['dist', 'node_modules'] },
10  js.configs.recommended,
11  ...tseslint.configs.recommended,
12  {
13    files: ['**/*.ts', '**/*.tsx'],
14    languageOptions: {
15      ecmaVersion: 2023,
16      globals: globals.browser,
17    },
18    plugins: {
19      'react-hooks': reactHooks,
20      'react-refresh': reactRefresh,
21    },
22    rules: {
23      ...reactHooks.configs.recommended.rules,
24      'react-refresh/only-export-components': ['warn', { allowConstantExport: true }],
25      '@typescript-eslint/no-unused-vars': ['error', { argsIgnorePattern: '^_' }],
26      '@typescript-eslint/consistent-type-imports': 'error',
27    },
28  },
29  {
30    // The service worker runs in a worker scope, not a window scope.
31    files: ['public/sw.js'],
32    languageOptions: {
33      sourceType: 'script',
34      globals: { ...globals.serviceworker, ...globals.browser },
35    },
36  },

```

```
37  prettier,  
38  );
```

apps/web/.prettierrc.json

JSON · 8 LINES

```
1  {  
2    "semi": true,  
3    "singleQuote": true,  
4    "trailingComma": "all",  
5    "printWidth": 100,  
6    "tabWidth": 2,  
7    "arrowParens": "always"  
8  }
```

apps/web/.prettierignore

IGNORE · 2 LINES

```
1  dist  
2  node_modules
```

apps/web/.dockerignore

SH · 7 LINES

```
1  node_modules  
2  dist  
3  .env  
4  .env.*  
5  !.env.example  
6  *.log  
7  .DS_Store
```

apps/web/Dockerfile

SH · 17 LINES

```
1  # ---- build -----  
2  FROM node:24-alpine AS build  
3  WORKDIR /app  
4  
5  COPY package.json package-lock.json ./  
6  RUN npm ci  
7  
8  COPY . .  
9  RUN npm run build  
10  
11 # ---- runtime -----  
12 FROM nginx:1.27-alpine AS runtime  
13  
14 COPY --from=build /app/dist /usr/share/nginx/html  
15 COPY nginx.conf /etc/nginx/conf.d/default.conf  
16  
17 EXPOSE 80
```

apps/web/nginx.conf

NGINX · 29 LINES

```
1  server {  
2    listen 80;  
3    server_name _;  
4    root /usr/share/nginx/html;  
5  
6    gzip on;  
7    gzip_types text/css application/javascript application/json image/svg+xml;  
8    gzip_min_length 1024;  
9  
10   location /api/ {  
11     proxy_pass http://api:4000;  
12     proxy_http_version 1.1;  
13     proxy_set_header Host $host;
```



```

14     proxy_set_header X-Real-IP $remote_addr;
15     proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
16     proxy_set_header X-Forwarded-Proto $scheme;
17 }
18
19 # Assets cache.
20 location /assets/ {
21     expires 1y;
22     add_header Cache-Control "public, immutable";
23 }
24
25 # SPA routing.
26 location / {
27     try_files $uri $uri/ /index.html;
28 }
29 }

```

apps/web/vite.config.ts

TS · 24 LINES

```

1 import { defineConfig, loadEnv } from 'vite';
2 import react from '@vitejs/plugin-react';
3 import tailwindcss from '@tailwindcss/vite';
4
5 export default defineConfig(({ mode }) => {
6     const env = loadEnv(mode, process.cwd(), '');
7
8     // Dev and preview only. Docker uses nginx for this.
9     const apiProxyTarget = env.API_PROXY_TARGET || 'http://localhost:4000';
10
11     const proxy = {
12         '/api': {
13             target: apiProxyTarget,
14             changeOrigin: true,
15         },
16     };
17
18     return {
19         plugins: [react(), tailwindcss()],
20         server: { host: true, port: 5173, proxy },
21         preview: { host: true, port: 4173, proxy },
22         build: { outDir: 'dist', sourcemap: false },
23     };
24 });

```

apps/web/index.html

XML · 21 LINES

```

1 <!doctype html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8" />
5     <meta
6       name="viewport"
7       content="width=device-width, initial-scale=1.0, viewport-fit=cover, maximum-scale=5.0"
8     />
9     <meta name="theme-color" content="#0f172a" />
10    <meta
11      name="description"
12      content="Log, track and resolve fabric quality defects from the shop floor."
13    />
14    <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
15    <title>Quality Inspection Tracker</title>
16  </head>
17  <body>
18    <div id="root"></div>
19    <script type="module" src="/src/main.tsx"></script>
20  </body>

```

21 </html>

apps/web/src/main.tsx

TS · 13 LINES

```
1 import { StrictMode } from 'react';
2 import { createRoot } from 'react-dom/client';
3 import App from './App';
4 import './index.css';
5
6 const container = document.getElementById('root');
7 if (!container) throw new Error('#root element is missing from index.html');
8
9 createRoot(container).render(
10   <StrictMode>
11     <App />
12   </StrictMode>,
13 );
```

apps/web/src/App.tsx

TS · 68 LINES

```
1 import { BrowserRouter, Navigate, Route, Routes } from 'react-router';
2 import { QueryClient, QueryClientProvider } from '@tanstack/react-query';
3 import { AuthProvider } from './context/AuthProvider';
4 import { useAuth } from './hooks/useAuth';
5 import { AppLayout } from './components/AppLayout';
6 import { ListPage } from './pages/ListPage';
7 import { NewInspectionPage } from './pages/NewInspectionPage';
8 import { InspectionDetailPage } from './pages/InspectionDetailPage';
9 import { SummaryPage } from './pages/SummaryPage';
10 import { LoginPage } from './pages/LoginPage';
11 import { ApiError } from './lib/api';
12 import { Spinner } from './components/ui';
13
14 const queryClient = new QueryClient({
15   defaultOptions: {
16     queries: {
17       staleTime: 30_000,
18       refetchOnWindowFocus: true,
19       retry: (failureCount, error) => {
20         if (error instanceof ApiError && !error.isNetworkError) return false;
21         return failureCount < 2;
22       },
23     },
24     mutations: {
25       // Default 'online' pauses mutations when offline and never settles the
26       // promise, leaving the save button spinning. Fail fast instead.
27       networkMode: 'always',
28     },
29   },
30 });
31
32 function AuthGate() {
33   const { user, isReady } = useAuth();
34
35   if (!isReady) {
36     return (
37       <div className="flex min-h-dvh items-center justify-center">
38         <Spinner label="Starting" />
39       </div>
40     );
41   }
42
43   if (!user) return <LoginPage />;
44
45   return (
46     <Routes>
```

```

47     <Route element={<AppLayout />}>
48       <Route index element={<ListPage />} />
49       <Route path="new" element={<NewInspectionPage />} />
50       <Route path="inspections/:id" element={<InspectionDetailPage />} />
51       <Route path="summary" element={<SummaryPage />} />
52     </Route>
53     <Route path="*" element={<Navigate to="/" replace />} />
54   </Routes>
55 );
56 }
57
58 export default function App() {
59   return (
60     <QueryClientProvider client={queryClient}>
61       <AuthProvider>
62         <BrowserRouter>
63           <AuthGate />
64         </BrowserRouter>
65       </AuthProvider>
66     </QueryClientProvider>
67   );
68 }

```

apps/web/src/index.css

TXT · 38 LINES

```

1  @import 'tailwindcss';
2
3  @theme {
4    --font-sans:
5      ui-sans-serif, system-ui, -apple-system, 'Segoe UI', Roboto, 'Helvetica Neue', Arial, sans-serif;
6  }
7
8  @layer base {
9    html {
10      -webkit-text-size-adjust: 100%;
11    }
12
13    body {
14      @apply bg-slate-100 text-slate-900 antialiased;
15      overflow-x: hidden;
16    }
17
18    /* Below 16px iOS zooms the page on focus. */
19    input,
20    select,
21    textarea {
22      font-size: 16px;
23    }
24
25    :focus-visible {
26      @apply outline-2 outline-offset-2 outline-slate-900;
27    }
28  }
29
30  @layer utilities {
31    /* Clears the iPhone home indicator. */
32    .pb-safe {
33      padding-bottom: max(env(safe-area-inset-bottom), 0.5rem);
34    }
35    .mb-safe {
36      margin-bottom: max(env(safe-area-inset-bottom), 0.5rem);
37    }
38  }

```

```

1 <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 32 32">
2   <!-- Weave lines under an inspection magnifier. -->
3   <rect width="32" height="32" rx="7" fill="#0f172a" />
4   <g stroke="#94a3b8" stroke-width="1.6" stroke-linecap="round">
5     <path d="M6 11h20M6 16h20M6 21h20" />
6   </g>
7   <circle cx="19" cy="13" r="6" fill="none" stroke="#fb7185" stroke-width="2.4" />
8   <path d="M23.4 17.4 27 21" stroke="#fb7185" stroke-width="2.4" stroke-linecap="round" />
9 </svg>

```

```

1 /// <reference types="vite/client" />
2
3 interface ImportMetaEnv {
4   readonly VITE_API_BASE_URL?: string;
5 }
6
7 interface ImportMeta {
8   readonly env: ImportMetaEnv;
9 }

```

```

1 import { NavLink, Outlet, useLocation } from 'react-router';
2 import { useAuth } from '../hooks/useAuth';
3 import { cn } from '../lib/cn';
4
5 const NAV_ITEMS = [
6   { to: '/', label: 'Inspections', icon: ListIcon, end: true },
7   { to: '/summary', label: 'Summary', icon: ChartIcon, end: false },
8 ];
9
10 export function AppLayout() {
11   const { user, logout } = useAuth();
12   const location = useLocation();
13
14   return (
15     <div className="flex min-h-dvh flex-col">
16       <header className="sticky top-0 z-20 bg-slate-900 text-white">
17         <div className="mx-auto flex max-w-lg items-center justify-between gap-3 px-4 py-3">
18           <div className="min-w-0">
19             <p className="truncate text-sm font-semibold leading-tight">Quality Inspection</p>
20             <p className="truncate text-xs text-slate-400">
21               {user ? user.displayName : 'Shop-floor tracker'}
22             </p>
23           </div>
24           {user && (
25             <button
26               type="button"
27               onClick={logout}
28               className="-mr-2 inline-flex min-h-11 shrink-0 items-center rounded-lg px-3 text-xs font-
semibold text-slate-300 hover:bg-slate-800"
29             >
30               Sign out
31             </button>
32           )}
33         </div>
34       </header>
35
36       <main className="mx-auto w-full max-w-lg flex-1 px-4 pt-4 pb-28">
37         <Outlet />
38       </main>
39     </div>

```

```

40     <nav
41       aria-label="Main"
42       className="pb-safe fixed inset-x-0 bottom-0 z-20 border-t border-slate-200 bg-white/95 backdrop-
blur"
43     >
44       <div className="mx-auto flex max-w-lg">
45         {NAV_ITEMS.map(({ to, label, icon: Icon, end }) => {
46           const active = end ? location.pathname === to : location.pathname.startsWith(to);
47           return (
48             <NavLink
49               key={to}
50               to={to}
51               end={end}
52               className={cn(
53                 'flex flex-1 flex-col items-center gap-0.5 px-2 pt-2 pb-1 text-xs font-medium',
54                 active ? 'text-slate-900' : 'text-slate-500',
55               )}
56             >
57               <Icon active={active} />
58               {label}
59             </NavLink>
60           );
61         })}
62       </div>
63     </nav>
64   </div>
65 );
66 }
67
68 function ListIcon({ active }: { active: boolean }) {
69   return (
70     <svg
71       width="22"
72       height="22"
73       viewBox="0 0 24 24"
74       fill="none"
75       stroke="currentColor"
76       strokeWidth={active ? 2.2 : 1.7}
77       strokeLinecap="round"
78       aria-hidden="true"
79     >
80       <path d="M8 6h12M8 12h12M8 18h12M3.5 6h.01M3.5 12h.01M3.5 18h.01" />
81     </svg>
82   );
83 }
84
85 function ChartIcon({ active }: { active: boolean }) {
86   return (
87     <svg
88       width="22"
89       height="22"
90       viewBox="0 0 24 24"
91       fill="none"
92       stroke="currentColor"
93       strokeWidth={active ? 2.2 : 1.7}
94       strokeLinecap="round"
95       strokeLinejoin="round"
96       aria-hidden="true"
97     >
98       <path d="M4 20V10M10 20V4M16 20v-7M22 20H2" />
99     </svg>
100   );
101 }

```

```

1  import { useEffect, useState } from 'react';
2  import { Button, Field, Input, Select } from './ui';
3  import { DEFECT_TYPE_LABELS, SEVERITY_LABELS, STATUS_LABELS } from '../lib/labels';
4  import { DEFECT_TYPES, SEVERITIES, STATUSES, type InspectionFilters } from '../lib/types';
5
6  export interface FilterSheetProps {
7    open: boolean;
8    value: InspectionFilters;
9    onClose: () => void;
10   onApply: (filters: InspectionFilters) => void;
11 }
12
13 export function FilterSheet({ open, value, onClose, onApply }: FilterSheetProps) {
14   if (!open) return null;
15   return <FilterSheetContent value={value} onClose={onClose} onApply={onApply} />;
16 }
17
18 function FilterSheetContent({ value, onClose, onApply }: Omit<FilterSheetProps, 'open'>) {
19   const [draft, setDraft] = useState<InspectionFilters>(value);
20
21   // Lock the body so the sheet scrolls, not the page.
22   useEffect(() => {
23     const onKey = (event: KeyboardEvent) => {
24       if (event.key === 'Escape') onClose();
25     };
26     window.addEventListener('keydown', onKey);
27     document.body.style.overflow = 'hidden';
28     return () => {
29       window.removeEventListener('keydown', onKey);
30       document.body.style.overflow = '';
31     };
32   }, [onClose]);
33
34   const set = <K extends keyof InspectionFilters>(key: K, raw: string) =>
35     setDraft((current) => ({
36       ...current,
37       [key]: raw === '' ? undefined : (raw as InspectionFilters[K]),
38       page: 1,
39     }));
40
41   const dateRangeInvalid = Boolean(draft.from && draft.to && draft.from > draft.to);
42
43   return (
44     <div className="fixed inset-0 z-40">
45       <button
46         type="button"
47         aria-label="Close filters"
48         onClick={onClose}
49         className="absolute inset-0 bg-slate-900/40"
50       />
51
52       <div
53         role="dialog"
54         aria-modal="true"
55         aria-label="Filter inspections"
56         className="pb-safe absolute inset-x-0 bottom-0 max-h-[88dvh] overflow-y-auto rounded-t-3xl bg-
white"
57       >
58         <div className="sticky top-0 bg-white px-4 pt-3 pb-2">
59           <div className="mx-auto mb-3 h-1 w-10 rounded-full bg-slate-300 aria-hidden="true" />
60           <h2 className="text-base font-semibold text-slate-900">Filters</h2>
61         </div>
62
63         <div className="flex flex-col gap-4 px-4 pb-4">
64           <Field label="Severity" htmlFor="filter-severity">

```

```

65     <Select
66       id="filter-severity"
67       value={draft.severity ?? ''}
68       onChange={(event) => set('severity', event.target.value)}
69     >
70       <option value="">All severities</option>
71       {SEVERITIES.map((severity) => (
72         <option key={severity} value={severity}>
73           {SEVERITY_LABELS[severity]}
74         </option>
75       ))}
76     </Select>
77 </Field>
78
79 <Field label="Status" htmlFor="filter-status">
80   <Select
81     id="filter-status"
82     value={draft.status ?? ''}
83     onChange={(event) => set('status', event.target.value)}
84   >
85     <option value="">All statuses</option>
86     {STATUSES.map((status) => (
87       <option key={status} value={status}>
88         {STATUS_LABELS[status]}
89       </option>
90     ))}
91   </Select>
92 </Field>
93
94 <Field label="Defect type" htmlFor="filter-defect">
95   <Select
96     id="filter-defect"
97     value={draft.defectType ?? ''}
98     onChange={(event) => set('defectType', event.target.value)}
99   >
100     <option value="">All defect types</option>
101     {DEFECT_TYPES.map((type) => (
102       <option key={type} value={type}>
103         {DEFECT_TYPE_LABELS[type]}
104       </option>
105     ))}
106   </Select>
107 </Field>
108
109 <Field label="Machine / line ID" htmlFor="filter-machine" hint="Partial match, e.g. LOOM">
110   <Input
111     id="filter-machine"
112     value={draft.machineId ?? ''}
113     onChange={(event) => set('machineId', event.target.value)}
114     placeholder="Any machine"
115     autoComplete="off"
116   />
117 </Field>
118
119 <div className="grid grid-cols-2 gap-3">
120   <Field label="From date" htmlFor="filter-from">
121     <Input
122       id="filter-from"
123       type="date"
124       value={draft.from ?? ''}
125       invalid={dateRangeInvalid}
126       onChange={(event) => set('from', event.target.value)}
127     />
128   </Field>
129   <Field
130     label="To date"
131     htmlFor="filter-to"
132     error={dateRangeInvalid ? 'Must be on or after the from date' : undefined}

```

```

133         >
134         <Input
135             id="filter-to"
136             type="date"
137             value={draft.to ?? ''}
138             invalid={dateRangeInvalid}
139             onChange={(event) => set('to', event.target.value)}
140         />
141     </Field>
142 </div>
143 </div>
144
145 <div className="pb-safe sticky bottom-0 flex gap-3 border-t border-slate-200 bg-white px-4 py-3">
146     <Button
147         variant="secondary"
148         fullWidth
149         onClick={() => {
150             const cleared: InspectionFilters = {
151                 sortBy: draft.sortBy,
152                 order: draft.order,
153                 page: 1,
154                 limit: draft.limit,
155             };
156             setDraft(cleared);
157             onApply(cleared);
158         }}
159     >
160         Clear all
161     </Button>
162     <Button fullWidth disabled={dateRangeInvalid} onClick={() => onApply(draft)}>
163         Apply
164     </Button>
165 </div>
166 </div>
167 </div>
168 );
169 }

```

apps/web/src/components/InspectionCard.tsx

TS · 55 LINES

```

1  import { Link } from 'react-router';
2  import { Badge } from './ui';
3  import {
4      DEFECT_TYPE_LABELS,
5      SEVERITY_BADGE,
6      SEVERITY_LABELS,
7      SEVERITY_RAIL,
8      STATUS_BADGE,
9      STATUS_LABELS,
10 } from '../lib/labels';
11 import { formatDate, relativeDay } from '../lib/format';
12 import type { Inspection } from '../lib/types';
13
14 export function InspectionCard({ inspection }: { inspection: Inspection }) {
15     return (
16         <Link
17             to={`/inspections/${inspection.id}`}
18             className="relative flex gap-3 overflow-hidden rounded-2xl bg-white p-3 pl-4 shadow-sm ring-1 ring-slate-200 active:bg-slate-50"
19         >
20             <span
21                 aria-hidden="true"
22                 className={`absolute inset-y-0 left-0 w-1.5 ${SEVERITY_RAIL[inspection.severity]} `}
23             />
24
25             <div className="min-w-0 flex-1">
26                 <div className="flex items-start justify-between gap-2">

```



```

27     <p className="truncate text-base font-semibold text-slate-900">{inspection.machineId}</p>
28     <Badge className={STATUS_BADGE[inspection.status]}>
29       {STATUS_LABELS[inspection.status]}
30     </Badge>
31   </div>
32
33   <p className="mt-0.5 truncate text-sm text-slate-600">
34     {DEFECT_TYPE_LABELS[inspection.defectType]}
35   </p>
36
37   <div className="mt-2 flex flex-wrap items-center gap-x-2 gap-y-1">
38     <Badge className={SEVERITY_BADGE[inspection.severity]}>
39       {SEVERITY_LABELS[inspection.severity]}
40     </Badge>
41     <span className="text-xs text-slate-500">
42       {formatDate(inspection.inspectionDate)} · {relativeDay(inspection.inspectionDate)}
43     </span>
44     {inspection.source === 'SAP_WEBHOOK' && (
45       <Badge className="bg-violet-100 text-violet-800 ring-violet-600/20">SAP</Badge>
46     )}
47   </div>
48
49   {inspection.remarks && (
50     <p className="mt-2 line-clamp-2 text-sm text-slate-500">{inspection.remarks}</p>
51   )}
52 </div>
53 </Link>
54 );
55 }

```

apps/web/src/context/AuthContextValue.ts

TS · 11 LINES

```

1 import { createContext } from 'react';
2 import type { AuthUser } from '../lib/types';
3
4 export interface AuthContextValue {
5   user: AuthUser | null;
6   isReady: boolean;
7   login: (username: string, password: string) => Promise<void>;
8   logout: () => void;
9 }
10
11 export const AuthContext = createContext<AuthContextValue | null>(null);

```

apps/web/src/context/AuthProvider.tsx

TS · 61 LINES

```

1 import { useCallback, useEffect, useMemo, useState, type ReactNode } from 'react';
2 import { AUTH_EXPIRED_EVENT, api, tokenStore } from '../lib/api';
3 import { AuthContext, type AuthContextValue } from './AuthContextValue';
4 import type { AuthUser } from '../lib/types';
5
6 export function AuthProvider({ children }: { children: ReactNode }) {
7   const [user, setUser] = useState<AuthUser | null>(null);
8   const [isReady, setIsReady] = useState(false);
9
10   useEffect(() => {
11     let cancelled = false;
12
13     void (async () => {
14       if (!tokenStore.get()) {
15         if (!cancelled) setIsReady(true);
16         return;
17       }
18
19       try {
20         const me = await api.me();

```

```

21     if (!cancelled) setUser(me.user);
22   } catch {
23     if (!cancelled) {
24       tokenStore.clear();
25       setUser(null);
26     }
27   } finally {
28     if (!cancelled) setIsReady(true);
29   }
30 }();
31
32 return () => {
33   cancelled = true;
34 };
35 }, []);
36
37 // Any 401 anywhere drops back to the login screen.
38 useEffect(() => {
39   const onExpired = () => setUser(null);
40   window.addEventListener(AUTH_EXPIRED_EVENT, onExpired);
41   return () => window.removeEventListener(AUTH_EXPIRED_EVENT, onExpired);
42 }, []);
43
44 const login = useCallback(async (username: string, password: string) => {
45   const result = await api.login(username, password);
46   tokenStore.set(result.token);
47   setUser(result.user);
48 }, []);
49
50 const logout = useCallback(() => {
51   tokenStore.clear();
52   setUser(null);
53 }, []);
54
55 const value = useMemo<AuthContextValue>(
56   () => ({ user, isReady, login, logout }),
57   [user, isReady, login, logout],
58 );
59
60 return <AuthContext.Provider value={value}>{children}</AuthContext.Provider>;
61 }

```

apps/web/src/hooks/useAuth.ts

TS · 8 LINES

```

1 import { useContext } from 'react';
2 import { AuthContext, type AuthContextValue } from '../context/AuthContextValue';
3
4 export function useAuth(): AuthContextValue {
5   const context = useContext(AuthContext);
6   if (!context) throw new Error('useAuth must be used inside <AuthProvider>');
7   return context;
8 }

```

apps/web/src/hooks/useInspections.ts

TS · 53 LINES

```

1 import { useMutation, useQuery, useQueryClient } from '@tanstack/react-query';
2 import { api } from '../lib/api';
3 import type { CreateInspectionInput, InspectionFilters } from '../lib/types';
4
5 export const inspectionKeys = {
6   all: ['inspections'] as const,
7   list: (filters: InspectionFilters) => ['inspections', 'list', filters] as const,
8   detail: (id: string) => ['inspections', 'detail', id] as const,
9   summary: ['inspections', 'summary'] as const,
10 };
11

```

```

12 export function useInspectionList(filters: InspectionFilters) {
13   return useQuery({
14     queryKey: inspectionKeys.list(filters),
15     queryFn: () => api.listInspections(filters),
16     placeholderData: (previous) => previous, // no flash of empty list while filtering
17   });
18 }
19
20 export function useInspection(id: string) {
21   return useQuery({
22     queryKey: inspectionKeys.detail(id),
23     queryFn: () => api.getInspection(id),
24     enabled: Boolean(id),
25   });
26 }
27
28 export function useSummary() {
29   return useQuery({ queryKey: inspectionKeys.summary, queryFn: api.getSummary });
30 }
31
32 export function useCreateInspection() {
33   const queryClient = useQueryClient();
34
35   return useMutation({
36     mutationFn: (input: CreateInspectionInput) => api.createInspection(input),
37     onSuccess: () => {
38       void queryClient.invalidateQueries({ queryKey: inspectionKeys.all });
39     },
40   });
41 }
42
43 export function useResolveInspection(id: string) {
44   const queryClient = useQueryClient();
45
46   return useMutation({
47     mutationFn: (resolutionNote: string) => api.resolveInspection(id, resolutionNote),
48     onSuccess: (updated) => {
49       queryClient.setQueryData(inspectionKeys.detail(id), updated);
50       void queryClient.invalidateQueries({ queryKey: inspectionKeys.all });
51     },
52   });
53 }

```

apps/web/src/lib/api.ts

TS · 117 LINES

```

1 import type {
2   AuthUser,
3   CreateInspectionInput,
4   Inspection,
5   InspectionFilters,
6   InspectionListResponse,
7   Summary,
8 } from './types';
9
10 // Relative by default: dev proxy and nginx both make the API same-origin.
11 const BASE_URL = import.meta.env.VITE_API_BASE_URL ?? '';
12
13 const TOKEN_KEY = 'qit.token';
14
15 export const tokenStore = {
16   get: () => localStorage.getItem(TOKEN_KEY),
17   set: (token: string) => localStorage.setItem(TOKEN_KEY, token),
18   clear: () => localStorage.removeItem(TOKEN_KEY),
19 };
20
21 export class ApiError extends Error {
22   readonly status: number;

```

```

23  readonly code: string;
24  readonly details?: unknown;
25
26  constructor(status: number, code: string, message: string, details?: unknown) {
27      super(message);
28      this.name = 'ApiError';
29      this.status = status;
30      this.code = code;
31      this.details = details;
32  }
33
34  // Request never reached the server.
35  get isNetworkError(): boolean {
36      return this.status === 0;
37  }
38
39  get fieldErrors(): Record<string, string[]> {
40      return this.details && typeof this.details === 'object'
41          ? (this.details as Record<string, string[]>)
42          : {};
43  }
44 }
45
46 export const AUTH_EXPIRED_EVENT = 'qit:auth-expired';
47
48 async function request<T>(path: string, init: RequestInit = {}): Promise<T> {
49     const token = tokenStore.get();
50     const headers = new Headers(init.headers);
51     if (init.body) headers.set('content-type', 'application/json');
52     if (token) headers.set('authorization', `Bearer ${token}`);
53
54     let response: Response;
55     try {
56         response = await fetch(`${BASE_URL}${path}`, { ...init, headers });
57     } catch {
58         throw new ApiError(0, 'NETWORK_ERROR', 'You appear to be offline.');

```

```

91 }
92
93 export const api = {
94   login: (username: string, password: string) =>
95     request<{ token: string; user: AuthUser }>('/api/auth/login', {
96       method: 'POST',
97       body: JSON.stringify({ username, password }),
98     }),
99
100   me: () => request<{ user: AuthUser | null }>('/api/auth/me'),
101
102   listInspections: (filters: InspectionFilters) =>
103     request<InspectionListResponse>(`/api/inspections${toQueryString(filters)}`),
104
105   getInspection: (id: string) => request<Inspection>(`/api/inspections/${id}`),
106
107   createInspection: (input: CreateInspectionInput) =>
108     request<Inspection>('/api/inspections', { method: 'POST', body: JSON.stringify(input) }),
109
110   resolveInspection: (id: string, resolutionNote: string) =>
111     request<Inspection>(`/api/inspections/${id}/resolve`, {
112       method: 'PATCH',
113       body: JSON.stringify({ resolutionNote }),
114     }),
115
116   getSummary: () => request<Summary>('/api/inspections/summary'),
117 };

```

apps/web/src/lib/cn.ts

TS · 3 LINES

```

1 export function cn(...classes: (string | false | null | undefined)[]): string {
2   return classes.filter(Boolean).join(' ');
3 }

```

apps/web/src/lib/format.ts

TS · 49 LINES

```

1 // Parsed as a calendar day so the date never shifts by timezone.
2 export function formatDate(dateOnly: string): string {
3   const [year, month, day] = dateOnly.split('-').map(Number);
4   if (!year || !month || !day) return dateOnly;
5   return new Date(Date.UTC(year, month - 1, day)).toLocaleDateString('en-GB', {
6     day: '2-digit',
7     month: 'short',
8     year: 'numeric',
9     timeZone: 'UTC',
10  });
11 }
12
13 export function formatDateTime(iso: string): string {
14   return new Date(iso).toLocaleString('en-GB', {
15     day: '2-digit',
16     month: 'short',
17     year: 'numeric',
18     hour: '2-digit',
19     minute: '2-digit',
20  });
21 }
22
23 // Today in the user's own timezone, the date they'd write down.
24 export function todayISO(): string {
25   const now = new Date();
26   const offsetMs = now.getTimezoneOffset() * 60_000;
27   return new Date(now.getTime() - offsetMs).toISOString().slice(0, 10);
28 }
29
30 export function daysAgoISO(days: number): string {

```

```

31  const now = new Date();
32  now.setDate(now.getDate() - days);
33  const offsetMs = now.getTimezoneOffset() * 60_000;
34  return new Date(now.getTime() - offsetMs).toISOString().slice(0, 10);
35 }
36
37 export function relativeDay(dateOnly: string): string {
38   const today = todayISO();
39   if (dateOnly === today) return 'Today';
40   if (dateOnly === daysAgoISO(1)) return 'Yesterday';
41
42   const diff = Math.round(
43     (Date.parse(`${today}T00:00:00Z`) - Date.parse(`${dateOnly}T00:00:00Z`)) / 86_400_000,
44   );
45   if (diff < 0) return 'Future date';
46   if (diff < 30) return `${diff} days ago`;
47   const months = Math.round(diff / 30);
48   return months <= 1 ? '1 month ago' : `${months} months ago`;
49 }

```

apps/web/src/lib/labels.ts

TS · 37 LINES

```

1  import type { DefectType, Severity, Status } from './types';
2
3  export const DEFECT_TYPE_LABELS: Record<DefectType, string> = {
4    WEAVE_DEFECT: 'Weave Defect',
5    SHADE_VARIATION: 'Shade Variation',
6    HOLE_TEAR: 'Hole / Tear',
7    COUNT_DEVIATION: 'Count Deviation',
8    OTHER: 'Other',
9  };
10
11 export const SEVERITY_LABELS: Record<Severity, string> = {
12   CRITICAL: 'Critical',
13   MAJOR: 'Major',
14   MINOR: 'Minor',
15 };
16
17 export const STATUS_LABELS: Record<Status, string> = {
18   OPEN: 'Open',
19   RESOLVED: 'Resolved',
20 };
21
22 export const SEVERITY_BADGE: Record<Severity, string> = {
23   CRITICAL: 'bg-rose-100 text-rose-800 ring-rose-600/20',
24   MAJOR: 'bg-amber-100 text-amber-900 ring-amber-600/25',
25   MINOR: 'bg-sky-100 text-sky-800 ring-sky-600/20',
26 };
27
28 export const SEVERITY_RAIL: Record<Severity, string> = {
29   CRITICAL: 'bg-rose-500',
30   MAJOR: 'bg-amber-500',
31   MINOR: 'bg-sky-500',
32 };
33
34 export const STATUS_BADGE: Record<Status, string> = {
35   OPEN: 'bg-slate-200 text-slate-800 ring-slate-500/20',
36   RESOLVED: 'bg-emerald-100 text-emerald-800 ring-emerald-600/20',
37 };

```

```
1  export const DEFECT_TYPES = [
2    'WEAVE_DEFECT',
3    'SHADE_VARIATION',
4    'HOLE_TEAR',
5    'COUNT_DEVIATION',
6    'OTHER',
7  ] as const;
8
9  export const SEVERITIES = ['CRITICAL', 'MAJOR', 'MINOR'] as const;
10 export const STATUSES = ['OPEN', 'RESOLVED'] as const;
11
12 export type DefectType = (typeof DEFECT_TYPES)[number];
13 export type Severity = (typeof SEVERITIES)[number];
14 export type Status = (typeof STATUSES)[number];
15 export type Source = 'MANUAL' | 'SAP_WEBHOOK';
16
17 export interface Inspection {
18   id: string;
19   inspectionDate: string;
20   machineId: string;
21   defectType: DefectType;
22   severity: Severity;
23   remarks: string | null;
24   status: Status;
25   resolutionNote: string | null;
26   resolvedAt: string | null;
27   source: Source;
28   clientRef: string | null;
29   createdAt: string;
30   updatedAt: string;
31 }
32
33 export interface Pagination {
34   page: number;
35   limit: number;
36   total: number;
37   totalPages: number;
38 }
39
40 export interface InspectionListResponse {
41   data: Inspection[];
42   pagination: Pagination;
43 }
44
45 export interface SeveritySummaryRow {
46   severity: Severity;
47   open: number;
48   resolved: number;
49   total: number;
50 }
51
52 export interface Summary {
53   totals: { open: number; resolved: number; total: number };
54   bySeverity: SeveritySummaryRow[];
55 }
56
57 export interface AuthUser {
58   sub: string;
59   username: string;
60   displayName: string;
61 }
62
63 export interface InspectionFilters {
64   status?: Status;
65   severity?: Severity;
```

```

66  defectType?: DefectType;
67  machineId?: string;
68  from?: string;
69  to?: string;
70  sortBy?: 'inspectionDate' | 'createdAt' | 'severity' | 'machineId';
71  order?: 'asc' | 'desc';
72  page?: number;
73  limit?: number;
74 }
75
76 export interface CreateInspectionInput {
77   inspectionDate: string;
78   machineId: string;
79   defectType: DefectType;
80   severity: Severity;
81   remarks?: string | null;
82 }

```

apps/web/src/pages/InspectionDetailPage.tsx

TS · 182 LINES

```

1  import { useState } from 'react';
2  import { Link, useNavigate, useParams } from 'react-router';
3  import { useInspection, useResolveInspection } from '../hooks/useInspections';
4  import { Badge, Button, Card, ErrorState, Field, Spinner, Textarea } from '../components/ui';
5  import {
6    DEFECT_TYPE_LABELS,
7    SEVERITY_BADGE,
8    SEVERITY_LABELS,
9    STATUS_BADGE,
10   STATUS_LABELS,
11 } from '../lib/labels';
12 import { formatDate, formatDateTime } from '../lib/format';
13
14 export function InspectionDetailPage() {
15   const { id = '' } = useParams();
16   const navigate = useNavigate();
17   const query = useInspection(id);
18   const resolve = useResolveInspection(id);
19
20   const [note, setNote] = useState('');
21   const [noteError, setNoteError] = useState<string | null>(null);
22   const [resolving, setResolving] = useState(false);
23
24   if (query.isLoading) {
25     return (
26       <div className="flex justify-center py-16">
27         <Spinner label="Loading inspection" />
28       </div>
29     );
30   }
31
32   if (query.isError || !query.data) {
33     return (
34       <div className="flex flex-col gap-3">
35         <ErrorState
36           message={query.error instanceof Error ? query.error.message : 'Inspection not found'}
37           onRetry={() => void query.refetch()}
38         />
39         <Link to="/">
40           <Button variant="secondary" fullWidth>
41             Back to inspections
42           </Button>
43         </Link>
44       </div>
45     );
46   }
47

```



```

48  const inspection = query.data;
49
50  const submitResolution = async () => {
51    const trimmed = note.trim();
52    if (trimmed.length === 0) {
53      setNoteError('A resolution note is required before closing this inspection.');

```

```

115     })
116   </dl>
117 </Card>
118
119 {inspection.status === 'RESOLVED' && inspection.resolutionNote && (
120   <Card className="border-l-4 border-emerald-500 p-4">
121     <h2 className="text-sm font-semibold text-slate-900">Resolution note</h2>
122     <p className="mt-1 text-sm whitespace-pre-wrap text-slate-700">
123       {inspection.resolutionNote}
124     </p>
125   </Card>
126 })
127
128 {inspection.status === 'OPEN' && !resolving && (
129   <Button fullWidth onClick={() => setResolving(true)}>
130     Mark as resolved
131   </Button>
132 })
133
134 {inspection.status === 'OPEN' && resolving && (
135   <Card className="flex flex-col gap-3 p-4">
136     <Field
137       label="Resolution note"
138       htmlFor="resolutionNote"
139       required
140       error={noteError ?? undefined}
141       hint="What was done to fix it? This is stored permanently."
142     >
143       <Textarea
144         id="resolutionNote"
145         rows={4}
146         autoFocus
147         value={note}
148         invalid={Boolean(noteError)}
149         onChange={(event) => setNote(event.target.value)}
150         placeholder="e.g. Reed replaced, loom re-qualified on a 50m trial run."
151       />
152     </Field>
153
154     <div className="flex gap-3">
155       <Button
156         variant="secondary"
157         fullWidth
158         onClick={() => {
159           setResolving(false);
160           setNoteError(null);
161         }}
162       >
163         Cancel
164       </Button>
165       <Button fullWidth disabled={resolve.isPending} onClick={() => void submitResolution()}>
166         {resolve.isPending ? 'Saving...' : 'Confirm resolved'}
167       </Button>
168     </div>
169   </Card>
170 })
171 </div>
172 );
173 }
174
175 function Row({ label, value }: { label: string; value: string }) {
176   return (
177     <div className="flex flex-col gap-0.5">
178       <dt className="text-xs font-medium tracking-wide text-slate-500 uppercase">{label}</dt>
179       <dd className="whitespace-pre-wrap text-slate-900">{value}</dd>
180     </div>
181   );

```

```

1 import { useMemo, useState } from 'react';
2 import { Link, useSearchParams } from 'react-router';
3 import { useInspectionList } from '../hooks/useInspections';
4 import { InspectionCard } from '../components/InspectionCard';
5 import { FilterSheet } from '../components/FilterSheet';
6 import { Badge, Button, EmptyState, ErrorState, Select, Spinner } from '../components/ui';
7 import { DEFECT_TYPE_LABELS, SEVERITY_LABELS, STATUS_LABELS } from '../lib/labels';
8 import { formatDate } from '../lib/format';
9 import type { InspectionFilters } from '../lib/types';
10
11 const SORT_OPTIONS = [
12   { value: 'inspectionDate:desc', label: 'Newest first' },
13   { value: 'inspectionDate:asc', label: 'Oldest first' },
14   { value: 'severity:asc', label: 'Most severe first' },
15   { value: 'severity:desc', label: 'Least severe first' },
16   { value: 'machineId:asc', label: 'Machine A-Z' },
17   { value: 'createdAt:desc', label: 'Recently logged' },
18 ] as const;
19
20 const PAGE_SIZE = 20;
21
22 // Filters live in the URL so views are shareable and back works properly.
23 function parseFilters(params: URLSearchParams): InspectionFilters {
24   const get = (key: string) => params.get(key) ?? undefined;
25   return {
26     status: get('status') as InspectionFilters['status'],
27     severity: get('severity') as InspectionFilters['severity'],
28     defectType: get('defectType') as InspectionFilters['defectType'],
29     machineId: get('machineId'),
30     from: get('from'),
31     to: get('to'),
32     sortBy: (get('sortBy') as InspectionFilters['sortBy']) ?? 'inspectionDate',
33     order: (get('order') as InspectionFilters['order']) ?? 'desc',
34     page: Number(get('page') ?? 1),
35     limit: PAGE_SIZE,
36   };
37 }
38
39 function toSearchParams(filters: InspectionFilters): URLSearchParams {
40   const params = new URLSearchParams();
41   for (const [key, value] of Object.entries(filters)) {
42     if (key === 'limit') continue;
43     if (value === undefined || value === null || value === '') continue;
44     if (key === 'page' && value === 1) continue;
45     params.set(key, String(value));
46   }
47   return params;
48 }
49
50 export function ListPage() {
51   const [searchParams, setSearchParams] = useSearchParams();
52   const [filterOpen, setFilterOpen] = useState(false);
53
54   const filters = useMemo(() => parseFilters(searchParams), [searchParams]);
55   const query = useInspectionList(filters);
56
57   const activeChips = buildChips(filters);
58   const update = (next: InspectionFilters) => setSearchParams(toSearchParams(next));
59
60   const pagination = query.data?.pagination;
61
62   return (
63     <div className="flex flex-col gap-3">

```

```

64 <div className="flex items-center gap-2">
65   <Button
66     variant="secondary"
67     onClick={() => setFilterOpen(true)}
68     className="shrink-0"
69     aria-haspopup="dialog"
70   >
71     <FilterIcon />
72     Filters
73     {activeChips.length > 0 && (
74       <span className="ml-0.5 rounded-full bg-slate-900 px-1.5 text-xs font-bold text-white">
75         {activeChips.length}
76       </span>
77     )}
78   </Button>
79
80   <label className="sr-only" htmlFor="sort">
81     Sort inspections
82   </label>
83   <Select
84     id="sort"
85     className="min-w-0 flex-1"
86     value={` ${filters.sortBy}: ${filters.order} `}
87     onChange={(event) => {
88       const [sortBy, order] = event.target.value.split(':');
89       update({
90         ...filters,
91         sortBy: sortBy as InspectionFilters['sortBy'],
92         order: order as InspectionFilters['order'],
93         page: 1,
94       });
95     }}
96   >
97     {SORT_OPTIONS.map((option) => (
98       <option key={option.value} value={option.value}>
99         {option.label}
100       </option>
101     ))}
102   </Select>
103 </div>
104
105 {activeChips.length > 0 && (
106   <div className="flex flex-wrap items-center gap-1.5">
107     {activeChips.map((chip) => (
108       <Badge key={chip.key} className="bg-slate-200 text-slate-800 ring-slate-400/30">
109         {chip.label}
110       </Badge>
111     ))}
112     <button
113       type="button"
114       onClick={() =>
115         update({ sortBy: filters.sortBy, order: filters.order, page: 1, limit: PAGE_SIZE })
116       }
117       className="rounded-full px-2 py-0.5 text-xs font-semibold text-slate-600 underline"
118     >
119       Clear
120     </button>
121   </div>
122 )}
123
124 <div className="flex items-center justify-between px-0.5">
125   <p className="text-xs text-slate-500">
126     {query.isLoading ? (
127       <Spinner label="Loading inspections" />
128     ) : pagination ? (
129       ` ${pagination.total} inspection${pagination.total === 1 ? '' : 's'} `
130     ) : null}
131   </p>

```

```

132     {query.isFetching && !query.isLoading && <Spinner label="Refreshing" />}
133   </div>
134
135   {query.isError && (
136     <ErrorState
137       message={
138         query.error instanceof Error ? query.error.message : 'Could not load inspections'
139       }
140       onRetry={() => void query.refetch()}
141     />
142   )}
143
144   {query.data && query.data.data.length === 0 && (
145     <EmptyState
146       title={activeChips.length > 0 ? 'No matching inspections' : 'No inspections yet'}
147       description={
148         activeChips.length > 0
149           ? 'Try widening the date range or clearing a filter.'
150           : 'Log the first defect you find on the floor and it will appear here.'
151       }
152       action={
153         <Link to="/new">
154           <Button>Log an inspection</Button>
155         </Link>
156       }
157     />
158   )}
159
160   <div className="flex flex-col gap-2">
161     {query.data?.data.map((inspection) => (
162       <InspectionCard key={inspection.id} inspection={inspection} />
163     ))}
164   </div>
165
166   {pagination && pagination.totalPages > 1 && (
167     <div className="mt-2 flex items-center justify-between gap-3">
168       <Button
169         variant="secondary"
170         disabled={pagination.page <= 1}
171         onClick={() => update({ ...filters, page: pagination.page - 1 })}
172       >
173         Previous
174       </Button>
175       <span className="text-xs text-slate-600">
176         Page {pagination.page} of {pagination.totalPages}
177       </span>
178       <Button
179         variant="secondary"
180         disabled={pagination.page >= pagination.totalPages}
181         onClick={() => update({ ...filters, page: pagination.page + 1 })}
182       >
183         Next
184       </Button>
185     </div>
186   )}
187
188   <Link
189     to="/new"
190     className="fixed right-4 bottom-20 z-30 inline-flex h-14 items-center gap-2 rounded-full bg-slate-
900 pr-5 pl-4 font-semibold text-white shadow-lg active:bg-slate-950"
191   >
192     <PlusIcon />
193     Log
194   </Link>
195
196   <FilterSheet
197     open={filterOpen}
198     value={filters}

```

```

199     onClose={() => setFilterOpen(false)}
200     onApply={next => {
201       update(next);
202       setFilterOpen(false);
203     }}
204   />
205 </div>
206 );
207 }
208
209 function buildChips(filters: InspectionFilters): { key: string; label: string }[] {
210   const chips: { key: string; label: string }[] = [];
211   if (filters.severity) chips.push({ key: 'severity', label: SEVERITY_LABELS[filters.severity] });
212   if (filters.status) chips.push({ key: 'status', label: STATUS_LABELS[filters.status] });
213   if (filters.defectType)
214     chips.push({ key: 'defectType', label: DEFECT_TYPE_LABELS[filters.defectType] });
215   if (filters.machineId) chips.push({ key: 'machineId', label: `Machine: ${filters.machineId}` });
216   if (filters.from || filters.to) {
217     const from = filters.from ? formatDate(filters.from) : 'Any';
218     const to = filters.to ? formatDate(filters.to) : 'Any';
219     chips.push({ key: 'dates', label: `${from} → ${to}` });
220   }
221   return chips;
222 }
223
224 function FilterIcon() {
225   return (
226     <svg
227       width="16"
228       height="16"
229       viewBox="0 0 24 24"
230       fill="none"
231       stroke="currentColor"
232       strokeWidth="2"
233       strokeLinecap="round"
234       aria-hidden="true"
235     >
236       <path d="M3 6h18M7 12h10M11 18h2" />
237     </svg>
238   );
239 }
240
241 function PlusIcon() {
242   return (
243     <svg
244       width="20"
245       height="20"
246       viewBox="0 0 24 24"
247       fill="none"
248       stroke="currentColor"
249       strokeWidth="2.5"
250       strokeLinecap="round"
251       aria-hidden="true"
252     >
253       <path d="M12 5v14M5 12h14" />
254     </svg>
255   );
256 }

```

```

1  import { useState, type SubmitEvent } from 'react';
2  import { useAuth } from '../hooks/useAuth';
3  import { Button, Card, Field, Input } from '../components/ui';
4
5  export function LoginPage() {
6    const { login } = useAuth();
7    const [username, setUsername] = useState('supervisor');
8    const [password, setPassword] = useState('');
9    const [error, setError] = useState<string | null>(null);
10   const [busy, setBusy] = useState(false);
11
12   const onSubmit = async (event: SubmitEvent) => {
13     event.preventDefault();
14     setBusy(true);
15     setError(null);
16     try {
17       await login(username, password);
18     } catch (caught) {
19       setError(caught instanceof Error ? caught.message : 'Could not sign in.');
```

```

66
67     <p className="mt-4 text-center text-xs text-slate-500">
68         Demo credentials – <span className="font-semibold">supervisor</span> /{' '}
69         <span className="font-semibold">arvind123</span>
70     </p>
71 </div>
72 </div>
73 );
74 }

```

apps/web/src/pages/NewInspectionPage.tsx

TS · 156 LINES

```

1  import { useState } from 'react';
2  import { useNavigate } from 'react-router';
3  import { useForm } from 'react-hook-form';
4  import { zodResolver } from '@hookform/resolvers/zod';
5  import { z } from 'zod';
6  import { useCreateInspection } from '../hooks/useInspections';
7  import { Button, ErrorState, Field, Input, Select, Textarea } from '../components/ui';
8  import { DEFECT_TYPE_LABELS, SEVERITY_LABELS } from '../lib/labels';
9  import { todayISO } from '../lib/format';
10 import { DEFECT_TYPES, SEVERITIES } from '../lib/types';
11
12 const formSchema = z.object({
13   inspectionDate: z
14     .string()
15     .min(1, 'Pick the date the defect was found')
16     .refine((value) => value <= todayISO(), 'The date cannot be in the future'),
17   machineId: z
18     .string()
19     .trim()
20     .min(1, 'Enter the machine or line ID')
21     .max(64, 'Keep this under 64 characters'),
22   defectType: z.enum(DEFECT_TYPES),
23   severity: z.enum(SEVERITIES),
24   remarks: z.string().trim().max(500, 'Keep remarks under 500 characters'),
25 });
26
27 type FormValues = z.infer<typeof formSchema>;
28
29 export function NewInspectionPage() {
30   const navigate = useNavigate();
31   const createInspection = useCreateInspection();
32   const [submitError, setSubmitError] = useState<string | null>(null);
33
34   const {
35     register,
36     handleSubmit,
37     formState: { errors, isSubmitting },
38   } = useForm<FormValues>({
39     resolver: zodResolver(formSchema),
40     defaultValues: {
41       inspectionDate: todayISO(),
42       machineId: '',
43       defectType: 'WEAVE_DEFECT',
44       severity: 'MAJOR',
45       remarks: '',
46     },
47   });
48
49   const onSubmit = handleSubmit(async (values) => {
50     setSubmitError(null);
51     try {
52       await createInspection.mutateAsync({
53         inspectionDate: values.inspectionDate,
54         machineId: values.machineId,
55         defectType: values.defectType,

```



```

56     severity: values.severity,
57     remarks: values.remarks.trim() === '' ? null : values.remarks.trim(),
58   });
59   navigate('/', { replace: true });
60 } catch (error) {
61   setSubmitError(error instanceof Error ? error.message : 'Could not save the inspection.');
```

```

62 }
63 });
64
65 return (
66   <form onSubmit={onSubmit} noValidate className="flex flex-col gap-4">
67     <div>
68       <h1 className="text-lg font-semibold text-slate-900">Log an inspection</h1>
69       <p className="mt-0.5 text-sm text-slate-500">
70         Record what you found on the floor. Fields marked * are required.
71       </p>
72     </div>
73
74     {submitError && <ErrorState message={submitError} />}
75
76     <Field
77       label="Date found"
78       htmlFor="inspectionDate"
79       required
80       error={errors.inspectionDate?.message}
81     >
82       <Input
83         id="inspectionDate"
84         type="date"
85         max={todayISO()}
86         invalid={Boolean(errors.inspectionDate)}
87         {...register('inspectionDate')}
88       />
89     </Field>
90
91     <Field
92       label="Machine / line ID"
93       htmlFor="machineId"
94       required
95       error={errors.machineId?.message}
96       hint="e.g. LOOM-14. Saved in uppercase."
97     >
98       <Input
99         id="machineId"
100         placeholder="LOOM-14"
101         autoCapitalize="characters"
102         autoComplete="off"
103         enterKeyHint="next"
104         invalid={Boolean(errors.machineId)}
105         {...register('machineId')}
106       />
107     </Field>
108
109     <Field label="Defect type" htmlFor="defectType" required error={errors.defectType?.message}>
110       <Select id="defectType" invalid={Boolean(errors.defectType)} {...register('defectType')}>
111         {DEFECT_TYPES.map((type) => (
112           <option key={type} value={type}>
113             {DEFECT_TYPE_LABELS[type]}
114           </option>
115         ))}
116       </Select>
117     </Field>
118
119     <fieldset className="flex flex-col gap-1.5">
120       <legend className="text-sm font-medium text-slate-700">
121         Severity<span className="ml-0.5 text-rose-600">*</span>
122       </legend>
123       <div className="grid grid-cols-3 gap-2">
```

```

124         {SEVERITIES.map((severity) => (
125             <label
126                 key={severity}
127                 className="relative flex min-h-11 cursor-pointer items-center justify-center rounded-xl bg-
white text-sm font-semibold text-slate-600 ring-1 ring-slate-300 has-checked:bg-slate-900 has-checked:text-
white has-checked:ring-slate-900"
128             >
129                 <input type="radio" value={severity} className="sr-only" {...register('severity')} />
130                 {SEVERITY_LABELS[severity]}
131             </label>
132         ))}
133     </div>
134 </fieldset>
135
136 <Field label="Remarks" htmlFor="remarks" error={errors.remarks?.message} hint="Optional">
137     <Textarea
138         id="remarks"
139         rows={3}
140         placeholder="What did you see? Where on the roll?"
141         invalid={Boolean(errors.remarks)}
142         {...register('remarks')}
143     />
144 </Field>
145
146 <div className="mt-2 flex gap-3">
147     <Button type="button" variant="secondary" fullWidth onClick={() => navigate(-1)}>
148         Cancel
149     </Button>
150     <Button type="submit" fullWidth disabled={isSubmitting}>
151         {isSubmitting ? 'Saving...' : 'Save inspection'}
152     </Button>
153 </div>
154 </form>
155 );
156 }

```

apps/web/src/pages/SummaryPage.tsx

TS · 108 LINES

```

1  import { Link } from 'react-router';
2  import { useSummary } from '../hooks/useInspections';
3  import { Button, Card, ErrorState, Spinner } from '../components/ui';
4  import { SEVERITY_LABELS, SEVERITY_RAIL } from '../lib/labels';
5
6  export function SummaryPage() {
7      const query = useSummary();
8
9      if (query.isLoading) {
10         return (
11             <div className="flex justify-center py-16">
12                 <Spinner label="Loading summary" />
13             </div>
14         );
15     }
16
17     if (query.isError || !query.data) {
18         return (
19             <ErrorState
20                 message={query.error instanceof Error ? query.error.message : 'Could not load the summary'}
21                 onRetry={() => void query.refetch()}
22             />
23         );
24     }
25
26     const { totals, bySeverity } = query.data;
27
28     return (
29         <div className="flex flex-col gap-4">

```

```

30 <div>
31   <h1 className="text-lg font-semibold text-slate-900">Summary</h1>
32   <p className="mt-0.5 text-sm text-slate-500">All inspections across the plant.</p>
33 </div>
34
35 <div className="grid grid-cols-3 gap-2">
36   <TotalTile label="Open" value={totals.open} tone="text-rose-700" />
37   <TotalTile label="Resolved" value={totals.resolved} tone="text-emerald-700" />
38   <TotalTile label="Total" value={totals.total} tone="text-slate-900" />
39 </div>
40
41 <section className="flex flex-col gap-2">
42   <h2 className="px-0.5 text-sm font-semibold text-slate-700">By severity</h2>
43
44   {bySeverity.map((row) => {
45     const resolvedPct = row.total === 0 ? 0 : Math.round((row.resolved / row.total) * 100);
46     return (
47       <Card key={row.severity} className="relative overflow-hidden p-4 pl-5">
48         <span
49           aria-hidden="true"
50           className={`absolute inset-y-0 left-0 w-1.5 ${SEVERITY_RAIL[row.severity]}`}
51         />
52
53         <div className="flex items-baseline justify-between gap-3">
54           <h3 className="text-base font-semibold text-slate-900">
55             {SEVERITY_LABELS[row.severity]}
56           </h3>
57           <span className="text-sm text-slate-500">{row.total} total</span>
58         </div>
59
60         <div className="mt-3 grid grid-cols-2 gap-3">
61           <Stat label="Open" value={row.open} tone="text-rose-700" />
62           <Stat label="Resolved" value={row.resolved} tone="text-emerald-700" />
63         </div>
64
65         <div className="mt-3">
66           <div
67             className="h-2 w-full overflow-hidden rounded-full bg-slate-200"
68             role="img"
69             aria-label={` ${resolvedPct}% of ${SEVERITY_LABELS[row.severity].toLowerCase()}
inspections resolved`}
70           >
71             <div
72               className="h-full rounded-full bg-emerald-500"
73               style={{ width: `${resolvedPct}%` }}
74             />
75           </div>
76           <p className="mt-1 text-xs text-slate-500">{resolvedPct}% resolved</p>
77         </div>
78       </Card>
79     );
80   })}
81 </section>
82
83 <Link to="/?status=OPEN&severity=CRITICAL&sortBy=inspectionDate&order=desc">
84   <Button variant="secondary" fullWidth>
85     View open critical inspections
86   </Button>
87 </Link>
88 </div>
89 );
90 }
91
92 function TotalTile({ label, value, tone }: { label: string; value: number; tone: string }) {
93   return (
94     <Card className="px-3 py-4 text-center">
95       <p className={`text-2xl font-bold tabular-nums ${tone}`}>{value}</p>
96       <p className="mt-0.5 text-xs font-medium text-slate-500">{label}</p>

```

```

97     </Card>
98   );
99 }
100
101 function Stat({ label, value, tone }: { label: string; value: number; tone: string }) {
102   return (
103     <div className="rounded-xl bg-slate-50 px-3 py-2">
104       <p className={`text-xl font-bold tabular-nums ${tone}`}>{value}</p>
105       <p className="text-xs font-medium text-slate-500">{label}</p>
106     </div>
107   );
108 }

```

apps/web/src/components/ui/index.tsx

TS · 202 LINES

```

1  import type {
2    ButtonHTMLAttributes,
3    InputHTMLAttributes,
4    ReactNode,
5    SelectHTMLAttributes,
6    TextareaHTMLAttributes,
7  } from 'react';
8  import { cn } from '../../lib/cn';
9
10 type ButtonProps = ButtonHTMLAttributes<HTMLButtonElement> & {
11   variant?: 'primary' | 'secondary' | 'ghost' | 'danger';
12   fullWidth?: boolean;
13 };
14
15 const BUTTON_VARIANTS = {
16   primary: 'bg-slate-900 text-white hover:bg-slate-800 active:bg-slate-950',
17   secondary: 'bg-white text-slate-900 ring-1 ring-slate-300 hover:bg-slate-50',
18   ghost: 'bg-transparent text-slate-700 hover:bg-slate-200',
19   danger: 'bg-rose-600 text-white hover:bg-rose-700',
20 } as const;
21
22 export function Button({
23   variant = 'primary',
24   fullWidth = false,
25   className,
26   ...props
27 }: ButtonProps) {
28   return (
29     <button
30       className={cn(
31         'inline-flex min-h-11 items-center justify-center gap-2 rounded-xl px-4 text-sm font-semibold',
32         'transition-colors disabled:cursor-not-allowed disabled:opacity-50',
33         BUTTON_VARIANTS[variant],
34         fullWidth && 'w-full',
35         className,
36       )}
37       {...props}
38     />
39   );
40 }
41
42 export function Field({
43   label,
44   htmlFor,
45   error,
46   hint,
47   required,
48   children,
49 }: {
50   label: string;
51   htmlFor: string;
52   error?: string;

```

```

53   hint?: string;
54   required?: boolean;
55   children: ReactNode;
56 }) {
57   return (
58     <div className="flex flex-col gap-1.5">
59       <label htmlFor={htmlFor} className="text-sm font-medium text-slate-700">
60         {label}
61         {required && (
62           <span className="ml-0.5 text-rose-600" aria-hidden="true">
63             *
64           </span>
65         )}
66       </label>
67       {children}
68       {hint && !error && <p className="text-xs text-slate-500">{hint}</p>}
69       {error && (
70         <p role="alert" className="text-xs font-medium text-rose-700">
71           {error}
72         </p>
73       )}
74     </div>
75   );
76 }
77
78 const CONTROL_BASE =
79   'min-h-11 w-full rounded-xl border-0 bg-white px-3 py-2 text-slate-900 ring-1 ring-slate-300 ' +
80   'placeholder:text-slate-400 focus:ring-2 focus:ring-slate-900 disabled:bg-slate-100';
81
82 export function Input({
83   invalid,
84   className,
85   ...props
86 }: InputHTMLAttributes<HTMLInputElement> & { invalid?: boolean }) {
87   return (
88     <input
89       aria-invalid={invalid || undefined}
90       className={cn(CONTROL_BASE, invalid && 'ring-rose-500 focus:ring-rose-600', className)}
91       {...props}
92     />
93   );
94 }
95
96 export function Select({
97   invalid,
98   className,
99   children,
100   ...props
101 }: SelectHTMLAttributes<HTMLSelectElement> & { invalid?: boolean }) {
102   return (
103     <select
104       aria-invalid={invalid || undefined}
105       className={cn(
106         CONTROL_BASE,
107         'appearance-none bg-no-repeat pr-9',
108         invalid && 'ring-rose-500',
109         className,
110       )}
111       style={{
112         backgroundImage:
113           "url(`data:image/svg+xml,%3csvg xmlns='http://www.w3.org/2000/svg' fill='none' viewBox='0 0 20 20'%3e%3cpath stroke='%2364748b' stroke-linecap='round' stroke-linejoin='round' stroke-width='1.5' d='M6 8l4 4-4-4/%3e%3c/svg%3e`)",
114         backgroundPosition: 'right 0.6rem center',
115         backgroundSize: '1.25rem',
116       }}
117       {...props}
118     >

```

```

119     {children}
120   </select>
121 );
122 }
123
124 export function Textarea({
125   invalid,
126   className,
127   ...props
128 }: TextareaHTMLAttributes<HTMLTextAreaElement> & { invalid?: boolean }) {
129   return (
130     <textarea
131       aria-invalid={invalid || undefined}
132       className={cn(CONTROL_BASE, 'min-h-24 resize-y', invalid && 'ring-rose-500', className)}
133       {...props}
134     />
135   );
136 }
137
138 export function Badge({ className, children }: { className?: string; children: ReactNode }) {
139   return (
140     <span
141       className={cn(
142         'inline-flex items-center rounded-full px-2 py-0.5 text-xs font-semibold ring-1 ring-inset',
143         className,
144       )}
145     >
146       {children}
147     </span>
148   );
149 }
150
151 export function Card({ className, children }: { className?: string; children: ReactNode }) {
152   return (
153     <div className={cn('rounded-2xl bg-white shadow-sm ring-1 ring-slate-200', className)}>
154       {children}
155     </div>
156   );
157 }
158
159 export function Spinner({ label = 'Loading' }: { label?: string }) {
160   return (
161     <span role="status" aria-label={label} className="inline-flex items-center gap-2">
162       <span className="h-4 w-4 animate-spin rounded-full border-2 border-slate-300 border-t-slate-900" />
163     </span>
164   );
165 }
166
167 export function EmptyState({
168   title,
169   description,
170   action,
171 }: {
172   title: string;
173   description: string;
174   action?: ReactNode;
175 }) {
176   return (
177     <div className="flex flex-col items-center gap-2 rounded-2xl bg-white px-6 py-12 text-center ring-1 ring-slate-200">
178       <p className="text-base font-semibold text-slate-900">{title}</p>
179       <p className="max-w-xs text-sm text-slate-500">{description}</p>
180       {action && <div className="mt-3">{action}</div>}
181     </div>
182   );
183 }
184
185 export function ErrorState({ message, onRetry }: { message: string; onRetry?: () => void }) {

```

```

186     return (
187       <div
188         role="alert"
189         className="flex flex-col items-start gap-3 rounded-2xl bg-rose-50 p-4 ring-1 ring-rose-200"
190       >
191         <div>
192           <p className="text-sm font-semibold text-rose-900">Something went wrong</p>
193           <p className="mt-0.5 text-sm text-rose-800">{message}</p>
194         </div>
195         {onRetry && (
196           <Button variant="secondary" onClick={onRetry}>
197             Try again
198           </Button>
199         )}
200       </div>
201     );
202   }

```